# STATEMENT OF WORK (SOW)

# LARGE COMPONENT BEAD BLAST BOOTH WITH PRE-ENGINEERED METAL BUILDING

**UNITED STATES AIR FORCE**
**OGDEN AIR LOGISTICS COMPLEX (OO-ALC)**

**09 MARCH 2026**

# 1. DESCRIPTION OF REQUIREMENT

1.1. Provide a complete and turn-key design, construction, and installation of a new corrosion control facility at the 309th Aerospace Maintenance and Regeneration Group (AMARG), located within a controlled area of Davis-Monthan Air Force Base (DMAFB), Arizona. The primary function is the corrosion control of Aerospace Ground Equipment (AGE) and other ferrous related items. The media blast booth will be housed within a fully enclosed Pre-Engineered Metal Building (PEMB) that also provides a separated maintenance support operations area (MX Ops Area). The entire facility shall be constructed on a currently unimproved site located North of Building 7213 (the Small Paint Booth), see appendix “A”.

1.2. The performing contractor is responsible for providing all management, professional services, labor, materials, and equipment to deliver a fully functional and operational facility. The scope includes, but is not limited to, the following key tasks:

1.2.1. Site Investigation and Design Development: Perform and bear all costs for all necessary preliminary and pre-construction investigations to develop the unimproved site. These investigations shall include, but are not limited to, geotechnical analysis, utility location and capacity assessment, environmental assessment, and topographical surveys. The Contractor shall be responsible for developing a complete design package, which shall be stamped and certified by the appropriate licensed Professional Engineer(s) registered in the State of Arizona.

1.2.2. Plastic Media Blast Booth: Furnish, install, and commission a complete, stand-alone system. The system must be appropriately sized for various AGE and include a full floor media recovery system, a dust collection system that meets all hazardous location requirements, and all associated controls and safety features.

1.2.3. PEMB: Design, furnish, and erect a new, permanent, weather-tight metal building to fully enclose the blast booth and provide a separate maintenance area. The structure must be fully compliant with all architectural and engineering standards outlined in the DMAFB Design Compatibility Guidelines.

1.2.4. Utilities and Site Work: Identify, design, and construct all supporting utility infrastructure; electrical, water, sanitary sewer, etc. This includes all site work, foundations, grading, drainage, and paving necessary to support the new facility.

1.3. GENERAL DESIGN REQUIREMENTS

1.3.1. Applicable guidance, regulations, & standards. As the Designer of Record, the Contractor is solely responsible for the research, identification, and compliance with all federal, state, local, and Department of Defense guidance, regulations, and standards. This includes, but is not limited to, all Unified Facilities Criteria (UFC), Air Force Instructions (AFI), National Fire Protection Association (NFPA) codes, and the Davis-Monthan AFB Design Compatibility Guidelines. The list provided in Appendix C is a baseline and is not exhaustive; the Contractor is responsible for adherence to all applicable regulations regardless of whether they are listed. In case of conflicting requirements, the most stringent regulation shall govern. The Contractor’s licensed Professional Engineer of Record shall assume full professional liability for the final constructed facility's compliance with all such governing criteria.

1.4. COTS PREFERENCE AND JUSTIFICATION

1.4.1. Requirement: The design shall prioritize and implement Commercial Off-The-Shelf (COTS) or modified-COTS solutions for all components and assemblies.

1.4.2. Non-COTS Justification and Analysis: Proposing a Non-COTS (custom fabricated) component or assembly is permissible only when it can be definitively proven that a COTS or Modified-COTS solution is incapable of meeting mandatory performance specifications. In such cases, the Contractor shall submit a formal Non-COTS Justification Report for each proposed item, demonstrating the following:

1.4.2.1. Analysis of Alternatives (AoA): Documented market research and analysis proving that no fewer than three (3) viable COTS or Modified-COTS alternatives can meet the government's minimum performance requirements as stated in this SOW. The AoA must detail why each alternative considered is technically insufficient.

1.4.2.2. Life Cycle Cost Analysis (LCCA): A comprehensive LCCA comparing the proposed Non-COTS item against the closest COTS alternative (even if deemed insufficient). The LCCA must be calculated over the 40-year engineered lifespan of the PEMB (as stated in 3.1.3) and shall include, at a minimum: initial acquisition and fabrication costs, installation, projected energy consumption, scheduled preventative.

1.4.3. The Contractor shall maintain a COTS Determination and Justification Log. This log must be submitted with each design stage (30%, 60%, 90%) and must accompany each Non-COTS Justification Report (CDRL A008) provided to the AMARG Contracting Officer (CO), Contracting Officer Representative (COR), and Project Manager(s) (PMs) for review and written approval prior to any procurement. The log shall include the minimum following fields:

1.4.3.1. Component Name

1.4.3.2. Required Performance Specification(s)

1.4.3.3. COTS Alternative(s) Considered (if any)

1.4.3.4. Justification for Non-COTS Selection (including specific reasons why COTS alternatives were unsuitable)

1.4.3.5. Cost Comparison

1.4.3.6. Approval: All Non-COTS selections require prior written approval from the CO.

1.5. CONTROL SYSTEM, INTERFACE, AND CYBERSECURITY REQUIREMENTS

1.5.1. System Software and Licensing. All software for the Programmable Logic Controller (PLC) and Human Machine Interface (HMI) shall utilize commercially supported platforms. The Contractor shall deliver all software, including the full developer/integrator suite, on physical media (e.g., USB drive, CD/DVD) along with all license keys. The Contractor shall provide a perpetual, transferable, enterprise-level license registered directly to the "309th AMARG, Davis-Monthan AFB," not to the contractor. The Government shall be provided all factory-default administrator-level credentials and permissions. If any software is custom developed it must be provided on physical media.

1.5.2. Human Machine Interface (HMI): The HMI shall provide unrestricted, real-time, read/write access to all operational setpoints, sensor readings, alarm thresholds, calibration tables, and control logic timers within the PLC. The HMI shall feature a Government-configurable, role-based access control system with, at a minimum, three tiers: Operator, Maintainer, and Administrator.

1.5.3. Cybersecurity and Authority to Operate (ATO), if applicable: The control system is a Facility-Related Control System and shall be designed, installed, and documented to achieve a formal Authority to Operate (ATO) from the cognizant Davis-Monthan AFB authorizing official. The Contractor shall be responsible for developing the full Risk Management Framework (RMF) package in accordance with UFC 4-010-06, Cybersecurity of Facility-Related Control Systems, and DoDI 8510.01. The system must meet, at a minimum, the security controls for Impact Level 2 (IL2), unless a higher level is specified by the Government.

1.5.3.1. The Hardware/Software list required in section 1.5.1 shall be delivered as a comprehensive Hardware/Software/Firmware Bill of Materials (BOM), formatted for use in the RMF process.

1.5.3.2. The Contractor shall perform all required security scans and remediate all findings at their own expense to ensure the system is capable of receiving an ATO prior to project turnover.

# 2. INVESTIGATION, PREPARATION, AND UTILITY DESIGN REQUIREMENTS

2.1. GENERAL SITE RESPONSIBILITY: The project site, located behind building 7213 as shown in Appendix A, is an unimproved area. The Contractor shall be solely responsible for all activities required to deliver a fully prepared and serviced site that meets all performance requirements of this SOW. This includes, but is not limited to, all investigation, design, engineering, permitting, and construction for site preparation and all utility systems.

2.2. DESIGN AND INVESTIGATION PREREQUISITES: As the initial phase of this turn-key effort, the Contractor, acting as the Designer of Record, shall perform and bear the full cost of all preliminary and pre-construction design services necessary to develop the unimproved project site. The Contractor is solely responsible for any unforeseen conditions that a comprehensive investigation would have revealed. These services shall include, but are not limited to:

2.2.1. A comprehensive topographical and boundary survey of the project site.

2.2.2. A complete geotechnical investigation, including soil borings, soil analysis, and providing foundation design recommendations in a formal report.

2.2.3. A thorough utility investigation to identify, locate (including depth), and assess the condition and available capacity of all existing utilities required for the facility (e.g., electrical, water, sanitary sewer, storm sewer, and communications). This includes identifying precise tie-in points.

2.2.4. An environmental site assessment to identify any potential environmental hazards or constraints.

2.2.5. Development of a comprehensive site plan that adheres to the DMAFB Design Compatibility Guidelines, detailing grading, drainage, utility routing, paving, force protection measures, and landscaping.

2.3. GEOTECHNICAL AND TOPOGRAPHICAL INVESTIGATION: The Contractor shall perform a comprehensive geotechnical investigation of the site in accordance with UFC 3-220-01, Geotechnical Engineering. The Contractor shall also perform a full topographical survey. The results of these investigations, including soil borings, soil classification, bearing capacity, and site grading requirements, shall be documented in a formal Site Investigation Report (SIR) (CDRL A002). The SIR shall be used as the basis of design for all foundations and earthwork and must be stamped by a licensed professional Civil or Geotechnical Engineer.

2.4. UTILITY SYSTEM DESIGN (CDRL A006): The Contractor shall conduct a comprehensive report of all existing base utility systems and determine their capacity to support the full build-out of this facility (i.e., the initial blast booth, the MX Ops Area, and the installation of a future blast booth of like size/capability within the MX Ops Area). This report shall include, at a minimum:

2.4.1. Electrical: Load testing of the nearest substation and feeders to verify capacity, voltage, phase, frequency, and amperage for the total project calculated load, and future expansion.

2.4.2. Water: Flow and pressure testing at the proposed tie-in point to verify adequacy for the required fire suppression systems and any process water needs.

2.4.3. Sewer: Investigation of existing sanitary sewer lines to determine capacity, invert elevations, and routing for the required connections.

2.4.4. Stormwater: Existing site topography and the location, capacity, and condition of the nearest storm drainage infrastructure (e.g., storm sewers, catch basins, retention ponds, or natural drainage channels).

2.5. UTILITY IMPROVEMENT DESIGN, VERIFICATION, AND COSTING: Based on Section 2.4 results, the Contractor shall design all necessary upgrades or new utility infrastructure. Any information or drawings of existing base utility systems provided by the Government are for reference purposes only. The Contractor is solely responsible for all field investigation and verification of existing utility conditions, including but not limited to locations, depths, sizes, materials, and available capacity at proposed points of connection. The final design shall extend from the verified point of connection to the facility. The cost for all identified and verified utility improvements shall be included as a firm-fixed price within the Contractor's total bid price. No claims for additional cost or time will be considered due to discrepancies between Government-provided reference drawings and actual field conditions.

2.5.1. The Contractor's firm-fixed-price shall be all-inclusive for delivering a fully serviced facility. This responsibility is absolute and not contingent upon the findings of the Contractor's own preliminary utility analysis. A failure by the Contractor to correctly forecast the required utility upgrades during the design phase shall not be considered a change in scope nor basis to claim additional compensation or time.

# 3. EXTERNAL STRUCTURE(S)

This section defines the requirements for the external structure or structures that will house the plastic media blast booth and associated facilities. The external structure shall be a Pre-Engineered Metal Building (PEMB) designed for modularity and ease of expansion.

## 3.1. GENERAL REQUIREMENTS

3.1.1. The external structure(s) shall meet all applicable building codes and Air Force design requirements.

3.1.2. Expandable: The PEMB shall be designed for modularity, allowing for future expansion with minimal disruption.

3.1.3. Durability: The PEMB shall be durable and resistant to weather, pests, and fire. The expected engineered lifespan of the PEMB shall be 40 years.

3.1.4. Aesthetics: The PEMB shall have an aesthetically pleasing appearance that is consistent with Davis Monthan Base Design Guide.

## 3.2. DIMENSIONS AND ORIENTATION

3.2.1. The PEMB shall be sized to accommodate the blast booth, all support/mechanical equipment, and the MX Ops Area as defined in Section 4 (Blast Booth and Supporting Equipment) & Section 5 (Maintenance Operations Area). The PEMB orientation shall ensure that the blast booth entrance door faces South (as shown in Appendix A).

3.2.2. All blasting equipment, including the blast booth and supporting equipment listed in Section 4 shall be consolidated and located on the East side of PEMB structure.

3.2.3. The MX Ops Area shall be on the west side of PEMB.

3.2.4. Demising Wall and Environmental Separation:

3.2.4.1. Fire Barrier: The blast operations area (East PEMB) and the MX Ops Area (West PEMB) shall be separated by a continuous, full height demising wall assembly. This assembly shall be designed and constructed as a 2-hour fire barrier in accordance with UFC 3-600-01 Fire Protection Engineering for Facilities, and the International Building Code (IBC).

3.2.4.2. Penetration Sealing: All penetrations through this fire barrier—including but not limited to conduits, pipes, and ductwork—shall be sealed with a UL-listed, 2-hour rated firestopping system appropriate for the specific type and size of the penetration.

3.2.4.3. In addition to the fire rating, this demising wall shall be designed and constructed as a continuous air barrier. To achieve this, all joints, seams, and junctions in the demising wall assembly—including but not limited to wall-to-floor, wall-to-wall, and wall-to-roof/deck connections—shall be continuously sealed on both sides with a permanent, flexible, non-hardening sealant prior to the installation of final finishes. The assembly shall achieve a maximum air leakage rate of 0.04 cfm/ft² at 75 Pa (1.57 psf) of pressure differential when tested. The Contractor shall be responsible for performing and documenting this pressure test across the barrier after installation of all services but before final finishes are applied. Any failure to meet the specified air leakage rate shall require the Contractor to identify and remediate all leak paths at their sole expense. The purpose of this requirement is to ensure no migration of hazardous dust or media from the "dirty" (East) side to the "clean" (West) side of the facility. The wall assembly's design must account for the positive pressurization of the clean spaces as specified in Sections 4.2.3 and 5.4.3.

3.3. MATERIALS

3.3.1. The PEMB shall be constructed of high-quality metal materials that are appropriate for the intended use and environment and meet all applicable code requirements.

3.3.2. The frame shall be constructed of structural steel.

3.3.3. The PEMB shall be insulated to meet climate control requirements in Section 4 and Section 5.

3.4. DOORS AND WINDOWS

3.4.1. The PEMB shall include all necessary doors and windows for access / ventilation and shall meet all applicable code requirements.

3.4.2. Equipment Doors: The PEMB shall include doors that are sized to ensure that the largest sized support equipment may be moved out of the structure for maintenance, replacement, or repair. The contractor shall provide dimensions of the largest piece of equipment that will be housed in this structure and ensure that exterior doors / openings are sized accordingly.

3.4.3. Doors: The PEMB shall include doors for personnel and equipment access in accordance with NFPA 101 life safety code and applicable occupancy group.

3.5. EXTERIOR FINISH

3.5.1. Requirement: The exterior of the PEMB shall be finished to provide a durable and aesthetically pleasing appearance that blends in with the surrounding structures.

3.5.2. Color: The main exterior color shall be consistent with the color of adjacent structures. Exterior color samples shall be submitted to 355 Civil Engineering for approval. There shall be exterior accents that match as closely as possible per DMAFB Building Design Guide; this includes gutters, downspouts, doors / framing. Color samples shall also be verified and approved by 355 Civil Engineering, prior to manufacturing and delivery.

3.5.3. Coatings: All structural steel shall be uniformly coated for corrosion resistance. Roof and wall panels shall be galvanized and painted.

3.6. STRUCTURAL DESIGN AND ENGINEERING

3.6.1. Designer of Record: The Contractor's structural engineer shall be the Structural Engineer of Record (SEOR) for the project. The SEOR shall be licensed in the state of Arizona and shall be responsible for the complete design and performance of the PEMB and its foundation.

3.6.2. Governing Standards and Design Loads: The structural design shall comply with all applicable codes and standards, with UFC 1-200-01, General Building Requirements, taking precedence. The design shall incorporate, at a minimum, the following criteria:

3.6.2.1. UFC 3-301-01, Structural Engineering: general structural design.

3.6.2.2. Wind Loads: Per UFC 3-301-01, calculated for a 120-mph nominal design wind speed, Exposure Category C, and a Risk Category III structure. The explicit use of "Aircraft Hangar Wind Loads" shall be evaluated by the SEOR for applicability and the most stringent condition applied.

3.6.2.3. Seismic Design: Site Soil Class D shall be used as a baseline, to be confirmed or revised by the Geotechnical Investigation required in Section 2.2 & 2.3.

3.6.2.4. Snow Load: 30 PSF Ground Snow Load, with drifts and unbalanced loads calculated per UFC 3-301-01 and ASCE 7-22.

3.6.3. Collateral and Suspended Loads: The structural design, including all frames, purlins, and girts, must support all specified collateral loads. As part of the 60% design submittal, the Contractor shall provide a Comprehensive Collateral Load Plan (CDRL A007). This plan shall include a scaled framing plan showing the precise location, weight, and structural attachment method for every piece of ceiling-mounted equipment, including but not limited to: fire suppression piping (filled with water), dry chemical fire suppression if applicable, fall arrest systems if applicable (including dynamic fall loads), HVAC units, all lighting fixtures, High Volume Low Speed (HVLS) fans, and any associated ductwork or cable trays.

3.6.4. PEMB Manufacturer Integration: The Contractor is responsible for ensuring that all loads and design requirements are fully integrated into the PEMB manufacturer's design and fabrication drawings (CDRL A001). All PEMB shop drawings must bear the stamp of the PEMB manufacturer's engineer as well as the review and approval stamp of the project's SEOR before submission to the government.

3.7. THERMAL PERFORMANCE

3.7.1. The PEMB insulation system shall be designed and installed to meet the prescriptive R-value or U-factor requirements for Climate Zone 2B (Tucson, AZ) as defined in ASHRAE 90.1 and UFC 1-200-02, High Performance and Sustainable Building Requirements. The insulation system shall ensure the HVAC systems can efficiently maintain the specified temperature ranges in Sections 4 and 5.

3.7.2. Insulation: To ensure long-term durability and prevent contamination from insulation fibers, a non-fibrous, puncture resistant, insulation system is preferred. Contractor to propose the most cost-effective system that meets this intent, while complying with IBC and NFPA standards.

3.8. PROVISIONS FOR FUTURE EXPANSION. The design shall incorporate specific, tangible provisions for future expansion. The Contractor shall identify the 'West' wall of the MX Ops Area as the 'Expansion Wall' in all design submittals.

3.8.1. Structural Provisions: The structural design of the Expansion Wall shall be engineered to be no-load-bearing regarding the primary building frame, allowing for the future removal of wall panels and girts without compromising the structural integrity of the initial facility.

3.8.2. Utility Provisions: The Contractor shall provide capped utility stub-outs, sized to support a duplicated facility load, at the Expansion Wall. This shall include, at a minimum, a capped water line for the future fire suppression system and appropriately sized, capped electrical conduit stub-outs from the main switchgear. The design must include isolation valves and breakers to allow for future tie-in without disruption to the initial facility.

3.8.3. Foundation Provisions: The foundation design shall extend beyond the Expansion Wall to form a footing for the future structure or shall be designed with engineered connection points (e.g., rebar dowels) to allow for a monolithic pour with the future foundation. All foundation drawings must clearly detail this provision.

3.8.4. Deliverables: The Contractor shall include a dedicated 'Future Expansion Plan' section within the 60% design submittal that details all structural, utility, and foundation provisions with calculations and drawings (CDRL A001).

3.9. PERSONNEL EGRESS

3.9.1. Provide exits or exit access doors around the entire perimeter of the PEMB to meet UFC 3-600-01, the NFPA 101 Life Safety Code and all other applicable personnel egress references.

3.9.2. Personnel egress shall be compliant with Americans with Disabilities Act (ADA) Standards for Accessible Design that apply to Government construction.

3.9.3. Exterior personnel doors shall be covered by awnings in accordance with OSHA 29 Code of Federal Regulations (CFR) 1910 and shall be designed for protection from falling objects and to prevent snow or ice accumulation. Exterior personnel doors shall be weatherproof and comprise of a minimum of 16-gauge metal with a U value of 0.50 or better.

3.9.4. Doors shall follow the keying and hardware requirements in Attachment F of the DMAFB Facility Design Guide, as applicable. Door cutouts shall be finished, framed, weatherproof, and corrosion resistant.

3.10. STORMWATER MANAGEMENT

3.10.1. Integrated Site Drainage Plan: Based on the investigation in Section 2, the Contractor shall design and construct a complete, integrated stormwater management system for the entire site of operations. This system shall comply with UFC 3-201-01, Civil Engineering, and all applicable base and local environmental regulations. The design shall, at a minimum:

3.10.1.1. Incorporate the building's roof drainage system (gutters and downspouts).

3.10.1.2. Include site grading, swales, catch basins, culverts, and any necessary underground piping to ensure positive drainage away from the facility foundation and prevent ponding or erosion anywhere on and off the site.

3.10.1.3. Manage stormwater for a 10-year storm event, at a minimum.

3.10.1.4. Include the development of a Stormwater Pollution Prevention Plan (SWPPP) for the construction phase, if required by local regulations.

3.10.2. The complete Stormwater Management design, including all calculations and grading plans, shall be stamped by the Contractor's licensed Professional Civil Engineer and included in the design submittals.

3.11. FIRE PROTECTION

3.11.1. The PEMB shall include a complete fire protection system compliant with all applicable codes, including UFC 3-600-01 and relevant NFPA standards. The Contractor shall perform and submit a formal Fire Protection Hazard Analysis as part of the 30% design submittal. This analysis will determine the specific system requirements (e.g., sprinkler type, coverage density, fire alarm integration). The final level and type of fire protection must be reviewed and approved in writing by the Davis-Monthan AFB Fire Department during the design review process. The Contractor is responsible for coordinating and documenting this approval.

3.11.2. Fire suppression system shall be approved by a licensed professional engineer, registered in any of the 50 United States as well as an NFPA Certified Fire Protection Specialist (CFPS).

3.11.3. The fire suppression system shall be tested IAW NFPA 409 Chapter 12 Section 12.5.9. Testing required by NFPA 409 Chapter 12 Section 11 shall also be performed as applicable to initial installation. Passing results shall be supplied to the GPE prior to acceptance.

3.12. LIGHTNING PROTECTION SYSTEM (LPS):

3.12.1. The LPS, if required, shall be designed and installed in accordance with AFMAN 32-1065, UFC 3-575-01, NEC, and NFPA 780.

3.12.2. Building ground systems shall be interconnected at a common accessible point in accordance with applicable regulations, including NFPA 780 Section 4.14.5 and Gap Analysis.

3.12.3. The system shall be installed by a contractor with an Underwriters Laboratories (UL) Lightning Protection Master Certification (LPMC).

3.12.4. The LPS design shall be approved by a licensed professional engineer, registered in any of the 50 United States.

3.12.5. Inspection and Acceptance:

3.12.5.1. The LPS be inspected prior to acceptance, by a commercial third-party inspector, whose sole work is lightning protection.

3.12.5.2. The system shall be certified by this third-party inspector in accordance with AFMAN 32-1065 and NFPA 780 (in that priority order).

3.13. ELECTRICAL

3.13.1. Contractor shall use appropriate NEMA rated enclosures for exterior and/or interior environments as applicable.

3.13.2. Provide an NEC compliant power interruption panel suitable for lock out/tag out IAW OSHA regulations for blast booth equipment.

3.14. LIGHTING

3.14.1. Internal overhead lighting shall be Solid-state lighting (SSL) Light Emitting Diode (LED), and installed IAW UFC 3-530-01, DMAFB Facility Design Guide and lighting levels per Illuminating Engineering Society (IES) recommendations with the following exceptions:

3.14.2. LED lighting shall be no more than 4100K at Color Rendering Index (CRI).

3.14.3. A photometric lighting plan shall be submitted to the GPE for approval through the design process (IAW para 6.3.1.).

3.14.4. Illumination of egress, emergency lighting, and egress markings shall be installed IAW NFPA 101 Section 40.2.8, 40.2.9, and 40.2.10.

3.14.5. Exit signs shall be the photo-luminescent type. Exit signs shall be connected to the emergency lighting inverter or provided with emergency battery units.

3.14.6. Interior lights shall be controlled via a combination of occupancy sensors, vacancy sensors, photocells, time clocks, and/or switches.

3.14.7. Exterior lighting to be installed IAW UFC 3-530-01, UFC 4-010-01, and DM Building Design Guide.

3.14.8. Exterior lighting shall be LED building-mounted fixtures installed over personnel egress. This includes any external utility or compressor rooms. No standalone light posts should be installed.

3.14.9. Exterior lighting shall be installed to illuminate the surrounding area and any disturbances to existing light coverage or shadowing caused by the PEMB.

# 4. BLAST BOOTH AND SUPPORTING EQUIPMENT (EAST PEMB)

This section defines the requirements for the blast booth support equipment / spaces of blast operations within the east side of the external structure, including locker rooms, transition area, observation room, and the equipment / storage room. All support facilities and spaces shall comply with applicable building codes, Air Force design requirements, and all applicable codes and regulations listed in Appendix C.

## 4.1. GENERAL REQUIREMENTS

4.1.1. Contamination Control Zone Designations: The entire blast operations facility shall be designed, constructed, and designated according to the three-zone "Cold/Warm/Hot" contamination control concept as detailed in UFC 4-211-02 figure 2-5 CLEAN-DIRTY Schematic (see Appendix “E”). The Contractor shall use this official nomenclature in all design documents, drawings, operational procedures, and signage. The zones are defined as follows:

4.1.1.1. Cold Zone (Clean Area): An area free from contamination where no Personal Protective Equipment (PPE) is required. This zone includes the Clean Locker Room, Administrative / Break Areas (if applicable), and the Observation Area. It shall be the area of highest relative air pressure.

4.1.1.2. Warm Zone (Contamination Reduction Corridor): A transition area between the Cold and Hot zones where personnel don off contaminated PPE. This zone includes the Dirty Locker Room, Respirator Cleaning Station, Transition Areas, and an Air Lock / Air Shower. It shall be maintained at a negative pressure relative to the Cold Zone.

4.1.1.3. Hot Zone (Exclusion Zone): This zone consists of the blast booth where hazardous operations occur and contamination is generated. Full PPE is required and shall be the area of most negative relative air pressure.

4.1.2. Layout: Support facilities and spaces shall be designed to eliminate the potential for blast booth media and dust migration and ensure proper workflow.

4.1.3. Finishes: All blast support and transition spaces shall have durable, easy-to-clean, wipeable, industrial level finishes. No carpet is permitted.

4.1.4. Fire Safety: The blast booth, mechanical areas, and support spaces shall include fire protection measures (e.g., fire sprinklers, fire-rated walls) as required by applicable codes and standards.

4.2. HVAC AND CONTAMINATION CONTROL

4.2.1. Climate Control: All ancillary spaces shall be climate controlled to maintain a target setpoint temperature between 60°F and 80°F.

4.2.2. Climate controls shall be localized / not connected to a network.

4.2.3. Contamination Control via Pressure Differential: The HVAC system shall be designed to create a cascading pressure differential to ensure airflow is always from cleaner areas to dirtier areas, preventing the migration of dust and contaminants. The design shall adhere to the principles outlined in UFC 4-211-02 "CLEAN-DIRTY" schematic, Blast-Resistant Design and the ACGIH Industrial Ventilation: A Manual of Recommended Practice. The pressure hierarchy shall be, at a minimum:

4.2.3.1. <u>Cold / Clean Spaces (Highest Pressure):</u> The Clean Locker Room and Observation Area shall be maintained at a positive pressure relative to all adjacent spaces.

4.2.3.2. <u>Warm / Transition Spaces (Intermediate Pressure):</u> The Dirty Locker Room and Transition Area shall be maintained at a negative pressure relative to the clean spaces, but a positive pressure relative to the blast enclosure.

4.2.3.3. <u>Hot / Dirty Spaces (Most Negative Pressure):</u> The blast enclosure itself shall be maintained at a negative pressure relative to all adjacent spaces, ensuring maximum containment.

4.2.4. Verification: The Contractor's certified Test and Balance reports must include a detailed matrix of differential pressure readings between all adjacent Cold, Warm, and Hot spaces to prove this cascading relationship has been achieved.

4.3. COLD, WARM, AND HOT AREAS

4.3.1. Configuration Approval. The Contractor shall be responsible for formally presenting the proposed layout of all Cold, Warm, and Hot areas to the Government's designated Bioenvironmental Engineering and Occupational Safety representatives during the 30% Concept Design stage. The Contractor shall document and incorporate all requirements from these stakeholders into the design. Written, signed approval of the final layout from both stakeholder offices must be obtained by the Contractor and submitted as a condition for the Government's approval of the 60% Preliminary Design submittal.

4.4. OBSERVATION ROOM (COLD AREA) SHALL:

4.4.1. Prevent entry or migration of blast dust.

4.4.2. Be sized to accommodate 2 personnel.

4.4.3. Have window sized for unobstructed view of the entire blast booth work area.

4.4.4. Have a window made of safety glass protected by a screen per OSHA 1910.94.

4.4.5. Be isolated from noise generated by the blast booth / equipment to a sustained level lower than 85 dBA during blast operations.

4.4.6. Provide access to all blast system and ventilation system controls and monitoring via the PLC/HMI.
4.4.7. Shall have full system emergency stop.
4.4.8. Shall have blast booth lighting controls.

4.5. CLEAN LOCKER ROOM (COLD AREA) SHALL:
4.5.1. Serve as the sole entrance into the Warm Area (Dirty Locker Room).
4.5.2. Be sized to accommodate a maximum of 4 personnel and have a minimum area of 200 square feet.
4.5.3. Contain four (4) lockers for storage of employees' personal items. Lockers will be stacked two high, lockable, with 18 inches of depth minimum.
4.5.4. Have Contractor installed supply cabinets for the storage of clean blast coveralls, gloves, and Tyvek suits.
4.5.5. Have two (2) seating benches, minimum length of 42 inches, and permanently mounted to flooring.

4.6. DIRTY LOCKER ROOM (WARM AREA) SHALL:
4.6.1. Be sized to accommodate a maximum of 4 personnel and have a minimum area of 200 square feet.
4.6.2. Contain four (4) lockers for storage of employees' personal items. Lockers will be stacked two high, lockable, with 18 inches of depth minimum.
4.6.3. Include space for government provided containers to collect contaminated PPE.
4.6.4. Have electrical outlets for use of a HEPA vacuum.
4.6.5. Have a culinary water source for a two basin sink that is sized large enough for hand and respirator washing. Basin sink will drain to the sanitary sewer, and all plumbing systems shall be installed IAW UFC 3-420-01.

4.7. AIR LOCK / AIR SHOWER (WARM AREA) SHALL:
4.7.1. Be provided for initial decontamination of employees who will transition to / from the blast booth (Hot Area) and Dirty Locker Room (Warm Area).
4.7.2. Be designed and implemented per NIOSH requirements and offer the best efficiency and efficacy of dust removal and filtration.
4.7.3. Be designed for ease of maintenance, specifically HEPA filter(s) changes.

4.8. TRANSITION AREA (WARM AREA) SHALL:
4.8.1. Have electrical outlets for HEPA vacuum.
4.8.2. Have indicator lights at booth entrance, showing status of blast enclosure (blast system enabled, hoses active/in use).

4.9. BLAST EQUIPMENT ROOM
4.9.1. An equipment area shall be provided for media reclaim, recovery, and blast equipment.
4.9.2. Blast equipment shall be designed to not leak or permit the migration of media or dust into the equipment area or Cold Areas.
4.9.3. External access via a roll-up door and a man door. Equipment / Storage doors shall be sized to ensure that the maintenance, replacement, or repair of the

equipment can be easily accomplished in and out of the room(s). The contractor shall provide dimensions of the largest piece of equipment that will be housed and ensure that the door is sized accordingly.

4.9.4. Shall have basic temperature control, ventilation, and/or insulation to prevent overheating (at or below 90°F) that is independent from ancillary personnel areas HVAC. Contractor to recommend most economical solution.

4.9.5. Shall have space to store 2 pallets of clean blast media barrels.

4.10. MECHANICAL/ELECTRICAL ROOM

4.10.1. A weatherproof compressor and electrical room shall be provided.

4.10.2. Size: The rooms shall be sized appropriately to accommodate all necessary electrical and compressor equipment.

4.10.3. Access: The rooms shall have secure access with lockable doors and provide external access for maintenance actions. Doors shall be sized to ensure that the maintenance, replacement, or repair of the equipment can be easily accomplished. The contractor shall provide dimensions of the largest piece of equipment that will be housed in this room and ensure that the door is sized accordingly.

4.11. COMPRESSED AIR SOURCE

4.11.1. The contractor shall be responsible for providing a compressed air source that is sufficient for the blast booth and all associated equipment for 8 hours of continuous blast booth operations.

4.11.2. Air needs to be dried to prevent issues with the blast media.

4.11.3. All compressed air tanks shall have auto drains installed to prevent buildup of condensate, oil, and/or corrosion. Contractor to recommend a permissible and most economical solution for drain condensate collection and/or disposal (i.e. sanitary sewer, other).

4.12. BREATHING AIR

4.12.1. The Contractor shall design, furnish, and install a complete, standalone, and compliant Grade D Breathing Air System. This system shall be designed in strict accordance with OSHA 29 CFR 1910.134, Respiratory Protection, and the referenced CGA G-7.1 standard. The system must be fully independent of the shop / process compressed air system.

4.12.2. Source and Filtration: The breathing air compressor shall be of a type specifically designed for respiratory protection (e.g., oil-less or with multi-stage filtration for oil removal). The compressor's air intake shall be located in a clean, outdoor area, demonstrably free from vehicle exhaust or other atmospheric contaminants. The system shall include all necessary filtration, drying, and purification stages to ensure the final air quality at the point of attachment meets or exceeds Grade D requirements under all operating conditions.

4.12.3. Continuous Monitoring and Alarms: The system shall be equipped with continuous in-line monitoring for Carbon Monoxide (CO). The CO monitor shall be the specified Invertech CAM-900. The system shall also incorporate alarms for high temperature and/or CO that are both audible and visual inside the blast booth and at the Observation Area HMI. The system must be configured to automatically shut down the breathing air supply and prevent use if a high-CO or high-temperature condition is detected.

4.12.4. Third-Party Verification and Certification: Upon completion of installation and prior to final acceptance, the Contractor shall engage a certified, independent, third-party testing laboratory to sample and test the breathing air from each connection point (CDRL A009). The Contractor shall provide the Government with a formal Certificate of Analysis from this laboratory certifying that the air delivered at each point meets or exceeds CGA G-7.1 Grade D specifications (CDRL A009). This testing and certification shall be performed at the Contractor's expense.

4.12.5. Breathing air couplings must not be compatible with outlets for no respirable worksite air or other gas systems. There should be multiple connection ports for airline respirator hoses to allow worker mobility.

4.13. ELECTRICAL

4.13.1. Sufficient to run all blast, recovery, reclaim, ventilation, and other equipment.

4.14. BLAST ENCLOSURE (Hot Area)

4.14.1. Shall comply with but not limited to T.O. 1-1-8, UFCs, all applicable building codes, Air Force design requirements, and applicable codes and regulations are listed in Appendix C.

4.14.2. Expected Workload for Blast Enclosure Table:

| AMARG AGE | | |
|---|---|---|
| Category | Equipment | Dimension |
| Tallest | B-2 Maintenance Stand | 17 ft lowered |
| Widest | B-52 Tow Bar | 12.5 ft |
| Heaviest | 4100 R& I Trailer | 3,190 lbs. |

4.14.3. Combustible Dust Hazard Compliance. The plastic media used in this facility is a recognized combustible dust. As such, the blast enclosure, dust collector, and all associated ventilation and reclaim ductwork shall be designed, installed, and protected in strict accordance with NFPA 652 Standard on the Fundamentals of Combustible Dust, and NFPA 654 Standard for the Prevention of Fire and Dust Explosions from the Manufacturing, Processing, and Handling of Combustible Particulate Solids. As a component of the 30% design submittal, the Contractor shall provide a formal Dust Hazard Analysis (DHA) as required by NFPA 652. This DHA shall inform the design of all required explosion protection and mitigation systems (e.g., explosion venting, suppression, or containment) which must be approved by the Davis-Monthan AFB Fire Department as part of the overall Fire Protection Hazard Analysis (per Section 3.11).

4.14.4. Floor Load Capacity: The blast booth floor shall be designed to support the weight of the heaviest configuration of AGE that will be used in the booth. The heaviest AGE is defined as the 4100 Engine Trailer (~3,190lbs).

4.14.4.1. Verification: The Contractor shall provide calculations demonstrating that the blast booth floor meets the specified load capacity. The calculations shall include the weight and footprint of the 4100 Engine Trailer.

4.14.5. Construction:

4.14.5.1. Structure will be delivered with most of the fabrication and assembly being accomplished at the factory and much of the installation shall be accomplished from the ground level.

4.14.5.2. May be integrated or an insert of the PEMB

4.14.6. Size:

4.14.6.1. The blast booth interior volume shall be sized to fit various sizes of Aerospace Ground Equipment that is listed in para 4.14.2.

4.14.6.2. Internal Dimensions:

4.14.6.2.1. Height: 20 ft

4.14.6.2.2. Width: 20 ft

4.14.6.2.3. Depth: 40 ft

4.14.6.3. Must be able to accommodate the following:

4.14.6.3.1. Tallest AGE: B-2 at 17’

4.14.6.3.2. Heaviest AGE: 3,190 lbs.

4.14.7. Walls for enclosure:

4.14.7.1. Sealed to prevent media/dust migration to surrounding areas

4.14.7.2. Walls need to be abrasion resistant

4.14.7.3. Sound damping material shall be installed on the exterior of blast booth walls to ensure noise levels are lower than 85 dBA during blast operations.

4.14.7.4. If integrated, then insulated appropriately

4.14.8. Lighting:

4.14.8.1. LED lighting on the walls and the ceiling appropriate for a blast environment, at a minimum Class 2 Div 1 or higher as required based on Fire Department classification of the space.

4.14.8.2. Minimum lighting is ~ 70-foot candles at 2’6” above the floor.

4.14.9. Utility Hookups:

4.14.9.1. Air connections should have a shut off valve at the wall connection point.

4.14.9.2. Air connections shall be managed by regulators to adjust pressure.

4.14.10. Compressed Air / Shop Air

4.14.10.1. 100 PSI.

4.14.10.2. Four (4) minimum connections.

4.14.10.3. Connection Type: DeVILBISS Quick Disconnect -High Flow Ball and Ring Lock Type.

4.14.11. Breathing Air Connections

4.14.11.1. Minimum of four (4) connections.

4.14.11.2. Breathing Air Couplets must not be compatible with outlets of non-breathing worksite air or other gases.

4.14.11.3. Connection Type: 3M Twist Lock (Shrader), Automatic Connect system.

4.14.11.4. Carbon Monoxide Monitor is required to ensure safe and compliant respirator program and shall be the Invertech CAM-900.

4.14.12. Notification System:

4.14.12.1. Breathing air system alarm.

4.14.12.2. Fire alarm.

4.14.12.3. Door alarm while in operation.

4.14.13. Doorways/Openings:

4.14.13.1. Sealed to prevent dust escape or migration.

4.14.14. Main Blast Door:

4.14.14.1. Sized to ensure fitment of tallest / widest AGE equipment.

4.14.14.2. Sealed to prevent dust escape, and ingress of outside moisture / dirt.

4.14.14.3. May be integrated in PEMB (preferred) or as an insert.

4.14.14.4. Door will be facing south towards wrap around road as shown in Appendix A.

4.14.15. Grounding:

4.14.15.1. Static Dissipation: The entire blast enclosure, including the operator and all equipment, shall be treated as a static-controlled area. The Contractor shall provide a complete static control system that includes, at a minimum conductive or static-dissipative blast hoses designed to prevent static charge accumulation.

4.14.15.2. Two (2) grounding points on / near centerline.

4.14.15.3. All blasting equipment and components being blasted shall be properly electrically grounded in accordance with TO 00-25-172.

4.14.15.4. Floor gates access panels should be located outside of the booth, where possible.

4.14.15.5. The blast booth must meet fire egress requirements per UFC 3-600-01.

4.14.15.6. Noise level within appropriate OSHA levels in and around the booth and support facilities & spaces.

4.15. BLAST EQUIPMENT

4.15.1. This section details the requirements for all equipment used in the plastic media blast process. The equipment shall be designed for maintainability and reliability, ensuring consistent and efficient performance. The blast equipment shall be capable of effectively removing paint, primer, and corrosion from AGE without damaging the underlying material. Blast booth shall utilize Type V media (Acrylic Plastic) with a 3.5MOH hardness defined in the Technical Order 1-1-8.

4.15.2. General Requirements:

4.15.2.1. All blast equipment shall be designed and constructed to meet all applicable safety standards.

4.15.2.2. Air Quality: Air shall be dried and filtered appropriately to prevent issues with the blast media or equipment.

4.15.2.3. Abrasion Resistance: All system transfer ducting shall be able to withstand abrasion from media.

4.15.2.4. Ergonomics and Noise Reduction: Blast equipment shall be designed with noise reduction technology and ergonomics to promote the reduction of user fatigue and safety.

4.15.3. Blast Equipment Requirements

4.15.3.1. System shall have two (2) independent blast pots capable of blasting continuously for 15 minutes with no interruptions.

4.15.3.2. The blast pots shall have capability to operate with media flow enabled or disabled (supplying compressed air only to the nozzle), selected for each pot independently from the Observation Room HMI or at the nozzle deadman trigger.

4.15.3.3. The blast pot pressure and the media feed rates shall be changeable for the individual blast pots, controlled from the Observation Room HMI using a supervisor-only pass code.

4.15.3.4. The blast pots shall be able to provide a calibrated and adjustable media flow rate between 0-15lbs/min with 0.1lb/min accuracy minimum, using an AccuFlow media metering valve, or equivalent.

4.15.3.5. The blast pots shall operate at a calibrated and adjustable pot pressure which provides 10-60 PSI at the blast nozzle with 2 PSI accuracy minimum.

4.15.4. Blast Hardware:

4.15.4.1. Nozzle- Marco #8 or equivalent.

4.15.4.2. Triggers – standard deadman trigger.

4.15.4.3. Media flow/air flow switch at nozzle.

4.15.4.4. Hoses (Main and whip hose), sized to reach the full interior space, with room to move vertically as needed.

4.15.4.5. Hoses shall be of a static-dissipative or conductive construction suitable for plastic media applications to prevent static charge buildup.

4.15.5. Media Reclaim Equipment

4.15.5.1. Requirement: Blast Booth shall utilize a pneumatic full floor media recovery system.

4.15.5.2. The media reclaim equipment shall be capable of efficiently and effectively reclaiming and cleaning used blast media for reuse.

4.15.5.3. Performance: The system shall be capable of processing and storing sufficient media to supply clean media to two (2) blast pots operating simultaneously for two (2) consecutive 8-hour shifts.

4.15.5.4. PLC Control: Equipment start/stop, appropriate adjustments, and monitoring shall be performed by the PLC/HMI in Observation Area.

4.15.5.5. Waste Management: The waste generated during the reclamation of the media shall be consolidated into a single waste stream into a waste collection system that is independent of the booth ventilation system. Waste shall be consolidated into a palletized 55-gallon open head steel drum with bolt ring.

4.15.5.6. System Transfer: All system transfer ducting should be able to withstand abrasion from the blast media.

4.15.6. Hopper Equipment Requirements (All):

4.15.6.1. Include a minimum of two (2) media storage hoppers including a dirty and clean hopper.

4.15.6.2. Include sensors to detect media capacities, to automate the start / stop of reclaiming and recovery equipment as needed, to keep the hoppers full. Hopper media levels shall be displayed in the Observation Room PLC.

4.15.6.3. Include sight windows to visually observe media levels.

4.15.6.4. Designed to prevent media bridging.

4.15.6.5. Designed to feed all blast pots evenly.

4.15.6.6. Self-feeding.

4.15.6.7. Include media shut off valves for maintenance.

4.15.7. Media Cleaning:

4.15.7.1. Timing adjusted from main HMI.

4.15.7.2. Ionizer to reduce static in media.

4.15.7.3. Vibratory classifier to sort media based on size.

4.15.7.4. Dense particle separator to sort media based on density.

4.15.7.5. Quality: The reclaim system shall clean media to meet the standards in TO 1-1-8, paragraph 2.12.12.

4.15.7.6. Anti-static fluid sprayer to treat reclaimed media (if required / optional)

4.15.7.6.1. Accessible refill point.

4.15.7.6.2. Applied only when the media is flowing.

4.15.7.6.3. Capability to adjust the amount of fluid sprayed.

4.15.7.7. Loading of new media; special reload mode to speed recovery of new media (floor gate control option)

4.16. BLAST BOOTH VENTILATION EQUIPMENT SHALL:

4.16.1. Provide a sufficient volume of ventilation air to maintain an average minimum cross-draft velocity of 100 FPM across the entire operational cross-section of the blast enclosure (OSHA 29 CFR 1910.94). The system shall be balanced to ensure airflow is laminar and without dead spots.

4.16.1.1. Ventilation Performance: Provide certified test and balance reports demonstrating that the blast enclosure meets the specified average cross-draft velocity of 100 FPM and the negative pressure requirements (per Section 4.2.3). Velocity readings shall be taken on a grid pattern as prescribed by the ACGIH Industrial Ventilation manual to verify uniform flow.

4.16.2. Produce negative pressure to minimize dust migration.

4.16.3. Maintain temperature below 80 in summer and above 60 in winter with ventilation system "on".

4.16.4. Clean air into front of booth, evacuate dirty/dusty air at the rear of the booth.

4.16.5. Industrial grade ducting suitable for airflow requirements.

4.17. DUST COLLECTORS:

4.17.1. Appropriate filters to remove dust from the air (generally main canister filters, pre-HEPA, and HEPA filters).

4.17.2. Differential Pressure (DP) readings across each filter bank independently and across the filter stack up Route DP readings to HMI screen.

4.17.3. Automatic canister filter cleaning via blow-offs (adjustable and monitored by DP readings).

4.17.4. Easy access to change filters.

4.17.5. Sealed against weather / water intrusion.

4.17.6. Screw auger to collect and transfer dust waste to 55-Gallon Open Head Steel Drum, bolt ring.

4.17.7. Explosion venting and protection per fire requirements.

4.17.8. Blower fans sized appropriately with automated Variable Frequency Drive (VFD) to provide continuous flow adjustments as needed.

4.17.9. Adjustable recirculation of dirty/clean air (with automated adjustments to meet temperature requirements).

## 5. MAINTENANCE OPERATIONS AREA (WEST PEMB)

This section details the requirements for the MX Ops Area located on the west end of the PEMB. This area will be designed to support current maintenance operations and the potential installation of a second blast booth of equal size / capability. This area shall be sealed to prevent migration of any blast booth contamination created in the East PEMB.

5.1. GENERAL REQUIREMENTS

5.1.1. Shall have a minimum square footage of 3600 square feet. This allows a duplication of the east blast booth system with a potential blast booth door facing on the west side of the PEMB.

5.1.2. With exception to life safety requirements, and to ensure designation as a COLD / Clean space by Bioenvironmental and Safety Representatives, the West PEMB shall be completely sealed off from East PEMB.

5.2. ELECTRICAL

5.2.1. Contractor shall provide 15 Amp 120v convenience outlets, at 10’ intervals, placed at equal distances along the north and south internal walls.

5.3. DOORS

5.3.1. A main exterior access door shall be placed on the most western end of the PEMB structure, with a minimum dimension door opening of 25’ tall and 20’ wide. Door shall be powered, with controls provided for opening and closing. Powered doors shall include all required safety devices as applicable.

5.4. HVAC

5.4.1. West PEMB shall have its own thermostat-controlled HVAC.

5.4.2. Target setpoint temperature to be 60 to 80 deg F unless otherwise noted.

5.4.3. Positively pressured to minimize dust migration from East PEMB.

5.4.4. Climate controls shall be localized / not connected to a network.

## 6. DESIGN

6.1. All design services furnished by the Design-Build Contractor shall be performed under the direct supervision of a licensed Professional Engineer. The Contractor, through its designer, is responsible for the accuracy, adequacy, timeliness and professionalism of the design solutions and design documents. The Contractor shall ensure design solutions meet requirements of this SOW and all governing regulations / standards.

6.2. The Government welcomes opportunities that will allow the schedule to be brought to the left or mitigate risk. If opportunities are identified, the Government is willing to consider for approval. i.e. procurement of long lead items during design development, manufacture of structural components, concurrent site preparations, etc.

6.3. Requirements include efficient project management including accurate, on-time submittals of contract deliverables and timely identification and solution of impediments to successful project execution. Technical requirements include early involvement in the process to allow for the development of the most cost- effective and technically sound solution. GPE will rely on the Contractor’s expertise in recognizing and addressing problematic issues and successful execution of this contract. The Contractor shall perform all work IAW federal, state, and local statutes and regulations.

6.4. Design Process and Schedule: The Contractor shall adhere to the following multi-stage design review process. Each submittal must be a complete, coordinated package incorporating all disciplines (Civil, Structural, Architectural, Mechanical, Electrical, Fire Protection). No physical construction or site mobilization shall occur until the Government provides written approval of the 100% "Issued for Construction" design and authorizes a formal Notice to Proceed (NTP) for construction. The timelines below represent the Contractor's work period, followed by the Government's review period. The Contractor shall not proceed to the next design stage without formal written comments or approval from the Government on the preceding stage.

6.4.1. Design Timeline Table (CDRL A001):

| DESIGN STAGES | DESIGN SCHEDULE TIMELINES |
|---|---|
| Concept Design (30%) | 21 Calendar Days from contract award |
| Government Review | 7 Calendar Days |
| Preliminary design submittal (60%) | 40 Calendar Days |
| Government Review | 10 Calendar Days |
| Pre-final design submittal (90%) | 40 Calendar Days |
| Government Review | 10 Calendar Days |
| Final Design Submittal | 21 Calendar Days |

6.5. Concept Design (30%) Submittal: This submittal establishes the basis of design. It shall include, at a minimum:

6.5.1. The complete Site Investigation Report (SIR).

6.5.2. Scaled floor plans and elevations showing spatial relationships, equipment layout, and personnel/equipment flow.

6.5.3. Single-line diagrams for all major mechanical and electrical systems.

6.6. Preliminary Design (60%) Submittal: This submittal develops the concepts into a defined engineering solution. It shall include, at a minimum:

6.6.1. All 30% deliverables, updated based on Government review comments.

6.6.2. Complete design calculations for all disciplines, stamped by the responsible professional engineer.

6.6.3. Drawings developed to show detailed floor plans, elevations, sections, and preliminary schedules (door, finish, equipment).

6.6.4. The Comprehensive Collateral Load Plan (as required by Section 3.6.3).

6.6.5. An initial draft of technical specifications, edited to identify applicable sections.

6.7. Pre-final Design (90%) Submittal: This submittal shall be a complete and final set of documents, intended for final government review before issuance. The "90% complete" designation reflects that it has not yet received final government sign-off. It shall include, at a minimum:

6.7.1. All 60% deliverables, updated and fully detailed.

6.7.2. Final, fully edited technical specifications.

6.7.3. Formal response log detailing resolution of all Government review comments.

6.7.4. A draft DD Form 1354 (CDRL A014); As the entity with direct knowledge of all final construction costs, the Contractor shall be responsible for preparing and submitting a complete draft DD Form 1354 that accurately categorizes all as-built costs in accordance with the data requirements of UFC 1-300-08, Criteria for Transfer and Acceptance of DoD Real Property, to facilitate the official transfer of the facility to the Government.

6.7.5. Contractor shall submit a comprehensive Commissioning and Acceptance Test Plan (CDRL A010). This plan shall detail the specific procedures, required equipment, responsible personnel, and quantifiable pass/fail criteria for every test required by this SOW. The plan must be approved by the GPE prior to the start of any commissioning activities

6.8. Final Design (100% / "Issued for Construction") Submittal: The Contractor shall incorporate all Government comments from the 90% review and resubmit the package. This submittal shall be stamped "Issued for Construction" by the Contractor's Designer of Record, signifying it is the approved set for construction. It must be accompanied by the final DD Form 1354.

6.8.1. Corrected final design submittal (100%) requirements. Unless otherwise approved the contractor shall not mobilize or start construction until the 100% (For Construction) submittal has been approved and construction Notice to Proceed has been authorized by the GPE. A final DD Form 1354 shall be completed and submitted with 100% design (CDRL A014).

6.8.2. Submittals: Civil Design, Structure and Trailer Construction drawings / details shall be submitted to the Contracting Officer Representative (COR) according to design schedule. Drawings shall be in .dxf, .pdf, and 3D .pdf formats (as applicable). Product literature shall be submitted in pdf. format or standard contractor format upon approval. No work shall proceed without approved submittals. Reviews will follow, at maximum, the schedule in SOW Para 6.4.1. If initial submittals are rejected, Contractor shall have 14 calendar days to resubmit from date of notification. The Government shall have 7 calendar days to review from date of receipt, unless otherwise negotiated. A design analysis shall be provided with design submittals incorporating all engineering computations, seismic analysis, test and survey results, building code review, fire protection analysis and calculations, structural load analysis and calculations, heating load analysis and calculations, energy code compliance documentation and all other similar pertinent information to clearly define the scope of the project and to express the designer's intent and methods. The author of each section shall include professional seals and signatures when appropriate.

6.8.3. As Built Drawings: Record Drawings shall be provided to the COR and OO-ALC/OBP within 30 days after completion of work (CDRL A011). Drawings must be complete, accurate and comply with current CAD standards.

6.8.4. The Government shall have the unrestricted right to use, modify, reproduce, release, perform, display, or disclose technical data associated with all designs, drawings, specifications, notes and other works developed and executed in the performance of this contract IAW DFARS 227.7102 – Commercial Items, Components or Processes and 252.227-7015 – Technical Data-Commercial Items. The Contractor for a period of three (3) years after acceptance of the final phase of the project agrees to furnish all retained works at the written request of the CO.

6.8.5. Commercial Off-The-Shelf (COTS) Equipment Operations and Maintenance (O&M) Manuals: All hard copy sets included with the equipment and original electronic copies shall be provided to the COR. Any modifications to COTS equipment shall be documented and included as part of the submission.

6.8.5.1. If readily available include the following:

6.8.5.1.1. Illustrated Parts Breakdowns

6.8.5.1.2. Exploded Views

6.8.5.1.3. Compatibility Literature

6.8.5.1.4. Troubleshooting Guides

6.8.5.1.5. Wiring Schematics

6.8.5.1.6. Maintenance Checklists

6.8.5.1.7. Performance Data

6.8.6. All equivalent or deviations from explicitly stated products shall be sent to the GPE for approval prior to installation.

6.9. Master Deliverables List: The Contractor shall provide all required data submittals in accordance with the project schedule. The following list represents the minimum required deliverables, which shall be formally tracked via a Contract Data Requirements List (CDRL) shown in para 6.9.7.1:

6.9.1. Design Submittals: (30%, 60%, 90%, 100%).

6.9.2. Site Investigation Report (SIR); Regulatory and Building Code Report; Basis of Design (BOD); Comprehensive Collateral Load Plan.

6.9.3. Cybersecurity Package (if applicable): Hardware/Software/Firmware Bill of Materials (BOM); Complete Risk Management Framework (RMF) package.

6.9.4. Verification & Test Documents: Non-COTS Justification Reports; Firestop Submittal and Installation Log; Air Barrier Test Report; Third-Party Breathing Air Certificate of Analysis; Commissioning and Acceptance Test Plan; Certified Test and Balance Reports.

6.9.5. Final Turnover Documents: Record "As-Built" Drawings; Comprehensive Operations & Maintenance (O&M) Manuals; Illustrated Parts Breakdowns; Complete COTS Manuals; Training Materials.

6.9.6. Administrative Documents: DD Form 1354 (Draft and Final).

6.9.7. CDRL Summary

6.9.7.1. Contract information data shall be delivered electronically in accordance with the corresponding requirement for each Data Item listed below. The contractor shall have no more than one error per quarter for all CDRL submittals. The contractor shall submit CDRLs according to the due date indicated and have no more than one CDRL per quarter submitted after the identified due date. Grammatical errors do not count as an error. Substantive mistakes shall be corrected within two working days and corrective action taken to prevent recurrence of inaccurate late CDRLs.

| Exhibit | Title | PWS Paragraph | Delivery Date |
|---|---|---|---|
| A001 | Design Submittals (30%, 60%, 90%, and 100%) | 3.6.4 & 3.8.4 & 6.4.1 | See Design Timeline Table |
| A002 | Site Investigation Report (SIR) | 2.3 | 30% Submittal |
| A003 | Accident/Incident Report TCIR/DART | 10.45.2 & Appendix D | Within three (3) calendar days of the accident/Incident |
| A004 | Quality/Safety Plan | Quality; 10.3.2<br><br>Safety Plan; 10.27.1 & 10.4.3 | Quality; No later than 30 days after contract award<br><br>Safety Plan; No later than 30 days after contract award |
| A005 | Program Progress Report | 10.29.1 | Provide on the second Monday of every month beginning the month after CO approval of the final design |
| A006 | Utility Systems Report | 2.4 | 30% Submittal |
| A007 | Comprehensive Collateral Load Plan | 3.6.3 | 60% Submittal |
| A008 | Non-COTS Justification Reports | 1.4.3 | 30%, 60%, 90% Submittals |
| A009 | Third-Party Breathing Air & Certificate of Analysis-Asking EN | 4.12.4 | Upon successful completion of pre-testing |
| A010 | Commissioning and | 6.7.5 & 7.1 | 90% Submittal |

| | | | |
|---|---|---|---|
| | Acceptance Test Plan | | |
| A011 | "As-Built" Drawings | 6.8.3 & 7.4 | Within 30 days of completion of work |
| A012 | Comprehensive Operations & Maintenance (O&M) Manuals | 7.4 | Final Acceptance |
| A013 | Training Materials | 8.3.1 | Contractor provided 30 days in advance of scheduled training |
| A014 | DD Form 1354 (Draft and Final). | Draft 6.7.4<br>Final 6.8.1 | Draft @ 90% Design<br>Final @ 100% Design |

# 7. SYSTEM COMMISSIONING AND ACCEPTANCE TESTING

7.1. Contractor Pre-Testing: The Contractor shall conduct all commissioning tests internally to verify and document that all systems are functioning correctly prior to the formal Government acceptance phase. This includes a minimum 80-hour operational run-in period with media flowing to identify and rectify any installation or equipment issues (per 1-1-8 para 2.12.12). (CDRL A010)

7.2. Formal Acceptance Testing: Upon successful completion of pre-testing, the Contractor shall formally notify the Government and schedule a multi-day Acceptance Testing and Demonstration event. All tests shall be witnessed by the GPE or their designated representative. The Contractor shall demonstrate, at a minimum:

7.2.1. System Calibration Verification: Demonstrate that nozzle pressure and media flow rates can be set at the HMI and are accurately delivered at the nozzle within the tolerances specified (per Section 4.15.3.5). This must be demonstrated at the minimum, median, and maximum operational ranges.

7.2.2. Safety System Verification: Demonstrate the functionality of every safety feature, including all emergency stops, alarms (per Section 4.14.12) , interlocks, and the breathing air system shutdown (per Section 4.12.3). This shall include a live CO monitor bump test.

7.2.3. Breathing Air Certification: Provide the final Certificate of Analysis from the third-party laboratory (as required by Section 4.12.4).

7.3. Operational Performance Demonstration: As the final stage of acceptance testing, the Contractor shall demonstrate the system's full operational capability. Government operators, using the contractor's provided training, will process three (3) different pieces of AGE (representative of the items in Table 4.14.2, to be provided by the Government) through a complete blast cycle. The system must perform without fault, and the results must meet the standards of T.O. 1-1-8. The contractor shall be responsible for addressing any issues identified during this demonstration.

7.4. Final Acceptance: The Government will grant final acceptance of the facility only after the successful completion of all witnessed tests, the successful Operational Performance Demonstration, and the receipt and approval of all required documentation and deliverables, including as-built drawings (CDRL A011) and O&M manuals (CDRL A012).

# 8. DOCUMENTATION & TRAINING REQUIREMENTS

8.1. Operations and Maintenance Manual encompassing blast booth and associated equipment to include air compressors, HVAC, doors, hardware, and structures.

8.1.1. Maintenance manuals will clearly state intervals of preventative maintenance and associated chemicals, lubrications, and equipment required to perform specified tasks.

8.2. Parts listing including manufacturer and model of all COTS items.

8.3. Training: The Contractor shall develop and provide a comprehensive training program for Government personnel on the operation, maintenance, and troubleshooting of the entire facility.

8.3.1. Training Materials: All training materials (manuals, presentations, guides) shall be submitted to the GPE for review and approval 30 days prior to the scheduled training (CDRL A013).

8.3.2. Training Event: The Contractor shall conduct on-site training for up to 8 operators and 4 maintainers.

8.3.3. Acceptance of Training: The training program will be deemed complete and accepted only after Government operators successfully and independently complete the Operational Performance Demonstration (as defined in Section 7.3), demonstrating full proficiency in operating the system.

8.4. Troubleshooting

8.4.1. Documentation on performing basic troubleshooting from an operator and maintainer level (i.e. nozzle adjustments, media recovery systems, others as identified by the Contractor).

# 9. SYSTEM WARRANTY AND SUSTAINMENT

9.1. Comprehensive Warranty. The Contractor shall provide a single, all-inclusive warranty for the entire turnkey facility for a period of one (1) year from the date of the Government's Final Acceptance (per Section 7.4). This warranty shall cover all materials, workmanship, and performance for every component of the project, with the explicit exception of the Pre-Engineered Metal Building (PEMB) structure, which is covered by the separate, longer-term warranty as defined in Section 9.2.

9.1.1. Individual equipment warranties shall be passed through from the original manufacturer’s standard warranty, with a minimum one (1) year warranty from date of final acceptance on all equipment and equipment workmanship.

9.2. PEMB Structural and Envelope Warranty. The Contractor shall provide a five (5) year warranty for the Pre-Engineered Metal Building (PEMB), effective from the date of final acceptance. This warranty shall cover, at a minimum, the structural integrity of all primary and secondary framing components, the weather-tightness of the complete building envelope (roof and walls), and the performance of all exterior coatings against peeling, cracking, or excessive fading.

9.3. Performance Guarantee: Under this warranty, the Contractor guarantees that every component of the project will continue to meet all performance requirements specified in this SOW. The Contractor shall be responsible for all costs associated with troubleshooting, parts, labor, and travel required to rectify any defect or failure within the warranty period, with corrective actions commencing within 72 hours of notification.

# 10. ADDITIONAL REQUIREMENTS

10.1. GOVERNING DOCUMENT NOMENCLATURE AND PRECEDENCE DURING ORGANIZATIONAL TRANSITION

10.1.1. The Contractor acknowledges that the Department of Defense (DoD) is in a period of transition to the Department of War (DoW). For the purposes of this contract, any and all references to 'DoD' within this SOW or any associated guidance, regulation, or standard shall be interpreted to mean 'DoW', and vice-versa. The Contractor is solely responsible for identifying and complying with the functionally relevant and currently effective publication, regardless of whether it is titled as a 'DoD' or 'DoW' document. In the event of a conflict between an older 'DoD' publication and a newly issued 'DoW' publication covering the same subject matter, the most recent publication shall take precedence

10.2. GOVERNMENT FURNISHED PROPERTY (GFP)

10.2.1. The Government shall provide access to the worksite, the place of performance shall be 309 AMARG, Tucson AZ.

10.2.2. If applicable, the Contractor shall be responsible for safeguarding all Government property provided for contractor use.

10.3. QUALITY MANAGEMENT

10.3.1. Contractor Quality Control (CQC). The Contractor shall establish and maintain a comprehensive Quality Control program to ensure all work, including the work of subcontractors and suppliers, complies with the requirements of this SOW. The Contractor is responsible for all quality control, and this responsibility is not relieved by any Government inspection or test.

10.3.2. Contractor Quality Control Plan (CQCP). The Contractor shall develop and submit a detailed Contractor Quality Control Plan (CQCP) for Government approval no later than 30 days after contract award (CDRL 004). No physical work may begin on site until the CQCP is formally approved by the Contracting Officer. The CQCP shall, at a minimum, detail the contractor's procedures for managing and controlling the quality of all design and construction activities, and must include:

10.3.2.1. The name and qualifications of the Contractor's designated Quality Control Manager (QCM), who shall report directly to the Contractor's on-site Project Manager and have the authority to stop work that does not conform to the SOW.

10.3.2.2. A list of all definable features of work.

10.3.2.3. Procedures for implementing the "Three Phases of Control" (Preparatory, Initial, and Follow-up inspections) for each definable feature of work.

10.3.2.4. A detailed testing and inspection schedule, identifying the tests required, the standards for those tests, and the organization responsible for conducting them.

10.3.2.5. Procedures for tracking and resolving all deficiencies from inception to correction.

10.3.2.6. A submittal management plan that aligns with the design schedule in Section 6.4.1.

10.3.3. Government Quality Assurance (QA). The Government will perform Quality Assurance inspections to verify that the Contractor's Quality Control program is functioning and that the work performed meets the requirements of the contract. The COR will inform the Contractor when discrepancies are noted. However, Government inspection does not relieve the Contractor of their sole responsibility to perform all work in accordance with the contract.

10.4. SAFETY AND HEALTH STANDARDS

10.4.1. The contractor shall follow and adhere to the Air Force and Occupational Safety and Health Administration (OSHA) standards.

10.4.2. The contractor shall provide from its OSHA 300 (Log of Work-Related Injuries and Illnesses) information the Total Case Incident Rate/Days Away Restricted or Transferred (TCIR/DART) rate by 15 January of each year to the CO for submission as part of the installation's annual OSHA VPP self-evaluation report.

10.4.3. System Safety Program Plan. The contractor shall implement a safety program that ensures protection of Government personnel and property. As part of the Safety Program, the contractor shall establish and submit a Safety Plan (also known as an Accident Prevention Plan or APP) IAW Appendix D of this SOW. This plan must be submitted to the Government for review and acceptance no later than 15 calendar days after contract award (CDRL 004). No notice to proceed for construction will be granted, and no physical work may begin on site, until the Safety Plan has been formally accepted by the Government.

10.5. MANPOWER REPORTING

10.5.1. The Contractor shall report ALL Contractor labor hours (including sub- contractor labor hours) required for performance of services provided under this contract via a secure data collection site. The Contractor is required to completely fill in all required data fields at http://www.SAM.gov

10.5.2. Reporting inputs will be for the labor executed during the period of performance for each Government fiscal year (FY), which runs 1 October through 30 September. While inputs may be reported any time during the FY, all data shall be reported no later than 31 October of each calendar year. Contractors may direct questions to the Contractor Manpower Reporting Application (CMRA) help desk.

10.5.3. Uses and Safeguarding of Information: Information from the secure web site is considered to be proprietary in nature when the contract number and Contractor identity are associated with the direct labor hours and direct labor dollars. At no time will any data be released to the public with the Contractor name and contract number associated with the data. User Manuals: Data for Air Force service requirements must be input at the ECMRA link. User manuals for Government personnel and Contractors are available at the ECMRA link at http://www.SAM.gov

10.6. DMAFB BASE ACCESS LONG-TERM PASSES (DBIDS CARD)

10.6.1. Passes for longer than 120 days may be picked up at 355th Security Forces Pass & ID (Craycroft Main Gate / Visitors Center), after the employee has been notified that the local files check has been completed during normal duty hours (0700 to 1600 M-F) by each individual with proper picture identification, proof of current insurance and registration for each vehicle. The contractor is responsible for requesting access to the base for their personnel by completing and submitting AF Form 496 to the COR or approved alternate. Long-term passes will be issued to the contract on-site staff. Information requirements are as follows:

10.6.1.1. Be in writing on company letterhead.

10.6.1.2. Be certified by an authorized representative of the prime Contractor that the specified individual(s)/vehicles require access to the base and will work on the project identified.

10.6.1.3. Specify the Contract Number.

10.6.1.4. Identify the specific company, (i.e. Contractor, subcontractor, supplier) covered by the request; each subcontractor request shall be separately submitted.

10.6.1.5. Specify the period of time when access will be required.

10.6.1.6. Identify specific personnel / vehicle information.

10.6.1.7. Provide the full name of each individual, including middle name and suffixes (e.g. Jr., III, etc.) Do not list aliases, nicknames, etc.

10.6.1.8. Provide each individual’s driver’s license number and state where issued; driver’s license must have ‘Real-ID’ endorsement / symbol.

10.6.1.9. Provide each individual’s social security number.

10.7. SHORT-TERM PASSES

10.7.1. Short-term passes are arranged through the Government by providing the name of the contractor, the contractor's driver's license number and state of issue. The Government will enter the contractors information into the 355th Security Forces Squadron (SFS) Visitor Control website at least 3 calendar days prior to the contractor's arrival for a local files check. The Contractor will need to have photo identification as well as a current driver's license, registration and proof of insurance if they intend to operate a vehicle on DMAFB.

10.8. UNAUTHORIZED ACTIVITY

10.8.1. The Contractor shall inform all personnel working under their jurisdiction (including subcontractor and supplier personnel) that access to areas outside of the immediate work area (excluding cafeterias and restroom near the work site, direct haul and access routes, Contracting and DMAFB Civil Engineering offices and points of supply and storage) is prohibited. Circulation of said personnel will be limited to official business only. Persons engaged in unauthorized reconnaissance of other contractor or Government activity will be referred to the CO for disposition. Infractions involving possible compromise of national security will be turned over to the Federal Bureau of Investigation (FBI) for disposition, and may result in termination of the contract.

10.9. ACCESS TO SURROUNDING BUILDINGS

10.9.1. Contractor shall provide 14-day notice prior to any disruption of government access to all buildings surrounding construction site. Duration for disruption shall be specified and approved by government prior to work being performed.

10.10. DENIAL OF BASE ACCESS

10.10.1. Access to DMAFB/AMARG and controlled areas can be denied based on criminal history and outstanding wants and warrants. If contractor personnel are not granted access to the installation based on criminal history and/or outstanding wants and warrants, the Contractor may be assessed liquidated damages IAW Federal Acquisition Regulation (FAR) 52.211-12 for any resultant schedule delay.

10.11. CONTRACTOR VEHICLES

10.11.1. The Contractor will be required to present proof of insurance, driver's license, vehicle registration, and other documentation as needed to operate a vehicle on DMAFB/AMARG. The company name shall be prominently displayed on both sides of all vehicles operated on DMAFB/AMARG.

10.12. SECURITY REQUIREMENTS

10.12.1. The contractor shall comply with all Security, Force Protection, and Antiterrorism requirements enforced at each installation. The contractor shall comply with local unit in / out processing procedures for all contractor personnel to ensure proper security-related briefings and debriefings are accomplished.

10.13. CONTRACTOR IDENTIFICATION BADGES AND ACCESS CARDS /

DBIDS CARDS

10.13.1. Contractor submits a completed Air Force Materiel Command (AFMC) Form 496, Application for AFMC Identification Card, to the Contracting Officer Representative (COR). The COR completes and forwards Form 496 to the Security Manager (SM) for signature and processing. The SM will notify the COR after Form 496 is approved with instructions for obtaining a DBIDS. Contractors shall return completed Form 496 to issuing SM to begin the in-processing. Contractor employees with issues associated with their information on their Form 496 will be disapproved for DBIDS and will not be granted access to the installation. NOTE: Escort authority will not be authorized, and Force Protection "Bravo" is the minimum established by Security Forces Squadron (SFS). Contractors with a current DBIDS have authority to sponsor (not escort) onto the installation and must follow installation procedures for doing so. Contractors who do not require access to DoD IT systems will be issued a DBIDS only.

10.13.2. Contractor employees approved for DBIDS will require sponsorship onto the installation (Craycroft Main Gate / Visitors Center)to be issued a DBIDS. Employees going to SFS without being sponsored are considered to be on the installation illegally.

10.13.2.1. Common issues with a denied Form 496: derogatory background, such as felony criminal charge/s (current or past), foreign national, missing information on form, or have failed to receive DBIDS within allotted time frame after approval.

10.13.2.2. If denied installation access, the employee will return all government issued credentials and will vacate the installation immediately. A separate notification will be provided by the 75 ABW/CC if an appeal is requested for access.

10.13.3. Controlled Area Badge (CAB) Process

10.13.3.1. Employees without a current DBIDS, or denied access to the installation, will not be issued a CAB until all issues are resolved.

10.13.3.2. The contractor shall provide, through the COR, written justification with employee full name, Height, Weight, Eye Color, Hair Color, contract number with expiration date and areas needing access to the issuing SM to initiate a request for a badge.

10.13.3.3. CAB requests shall be completed by the group SM, who shall complete an AF Form 2586, Unescorted Entry Authorization Certificate, for each employee requiring entry into controlled areas. The contractor employee shall carry the AF Form 2586 to Security Forces (building 430). The AF Form 2586 will be returned to the issuing SM for accountability. NOTE: CABs will not be issued for the inconvenience of having to be escorted.

10.13.4. While in the Controlled Areas, contractor employees shall display the CAB in a badge holder, furnished by the contractor company, between the shoulders and above the waist and in the front of the body whenever inside an identified controlled area. NOTE: When outside controlled areas the CAB will be placed out of sight.

10.13.5. Security Education and Motivation training shall be completed, initially and annually thereafter in order to obtain, and maintain, a CAB. Employees who fail to meet training requirements will have their credentials removed and installation access revoked.

10.13.6. Training can be expected to last one and one half to two hours long, initially and annually. Training is scheduled with the Group SM and requires a passing score to receive approval for a CAB.

10.14. VINDICATOR ACCESS CARD

10.14.1. As required, the contractor employee is issued a vindicator access card from the appropriate Building Facility/SM. Access to controlled areas will not be granted without a CAB badge and proper authorization for required areas. NOTE: CORs shall provide DBIDS expiration date, CAB number and expiration date, vindicator card number, four digit pin number and areas requested for access to the group SM prior to gaining access to a controlled area.

10.14.2. Lost Badges or Access Cards shall be reported to the Maintenance Support Group (MXSG) Security Office within one working day of loss. After receiving a Lost Identification Letter a new Form 2586 will be issued. The contractor must carry all forms to building 430 to be issued a new badge then return all forms to the MXSG Security Office. Due to unforeseen circumstances employees may be required to call (801) 586- 5199 to schedule an appointment.

10.14.3. The contractor shall create an Incident/Mishap report when a contract employee loses a Controlled Area Badge (Line badge). The Incident/Mishap Report must be submitted no later than one (1) business day.

10.14.4. Returning of credentials shall be done within two (2) working days when the Period of Performance (POP) has ended or the requirement is considered complete. During performance of the contract and at the completion or termination of the contract, the contractor shall be responsible for prompt return of credentials to the appropriate security manager for any employee who no longer requires access to the work site, and for obtaining required identification of newly assigned personnel.

10.15. SECURITY INCIDENTS

10.15.1. Should a security incident occur, the contractor shall:

10.15.1.1. Immediately report to the site location Group Security Manager all available facts related to each instance.

10.15.1.2. Take such precautions as the Security Manager may reasonably require for security purposes.

10.15.1.3. Take reasonable and prudent action to establish control of the scene, prevent further violation, and to preserve evidence until released by proper authority.

10.15.1.4. Cooperate fully and assist Government personnel as the Government conducts an investigation of the violation.

10.15.1.5. Participation shall last until the investigation is complete.

10.16. KEYS, BUILDING ACCESS CARDS, AND BUILDING BADGE CONTROL

10.16.1. The Contractor shall establish and implement methods of making sure that all keys, vindicator cards, and building badges issued to the Contractor by the Government are not lost, misplaced, or used by unauthorized persons.

10.16.2. The Contractor shall not duplicate any keys issued by the Government.

10.16.3. The Contractor shall immediately report the occurrence of a lost key to the appropriate building Facility Manager, and vindicator cards and building badges to the appropriate Security Managers.

10.16.4. The Contractor shall prohibit the opening of locked areas by the Contractor employees to permit entrance of persons other than the Contractor's employees engaged in the performance of assigned work in those areas, or personnel authorized entrance by the Security Managers.

10.17. COMMUNICATIONS SECURITY

10.17.1. All communications with DoD organizations are subject to Communications Security (COMSEC) review.

10.17.2. The DoD authorizes the military departments to conduct COMSEC monitoring and recording of telephone calls originating from, or terminating at, DoD organizations.

10.17.3. The Government advises the Contractor that at any time they place a call to, or receive a call from, a DoD organization, they are subject to COMSEC procedures.

10.17.4. The Contractor assumes the responsibility for ensuring wide and frequent dissemination of the above information to all employees dealing with official DoD information.

10.18. OPERATIONS SECURITY (OPSEC)

10.18.1. Public release of any information is not authorized and, if requested, will comply with Public Affairs and all other OPSEC measures to ensure the best balance of lawful disclosure for the American public and denial of advantage for our adversaries.

10.18.2. No photography, or videotaping, is allowed on the installation without approval from the owning organizations Group Director/Commander on an approved photography letter from Public Affairs. This includes the use of personal cell phones with picture and video capability.

10.18.3. Critical Items/Information shall not be discussed outside normal contract conditions. A Critical Items/ Information List (CIL) can be provided and should not be considered all inclusive.

10.19. Controlled Unclassified Information (CUI) generated by the contract, and its employees, shall ensure protective measures are applied. ALL DoD unclassified information MUST BE REVIEWED AND APPROVED FOR RELEASE before it's provided to the public in accordance with DoDD 5230.09, Reference J.

10.19.1. Disclosure of information is prohibited for release of unclassified information to the public without approval of the contracting activity.

10.19.2. Contractors will comply with CUI requirements detailed in DoDM 5200.01-V4, DoD Information Security Program: Controlled Unclassified Information (CUI).

10.20. SECURITY:

10.20.1. Contractor employees who have access to government facilities or critical information shall be trained on Operations Security (OPSEC), human relations, and security education and motivation within 90 days after initial assignment of the contract IAW AFI 10-701 Para. 4.1-4.2.

10.20.2. During performance of the contract and at the termination or completion of the contract, the contractor shall be responsible for obtaining required identification for newly assigned personnel, and for prompt return of credentials to the 309 AMARG Security Manager, 801-586- 4480, for any employee who no longer requires access to the work site. Lost badges shall be reported to the Maintenance Group Security Officer within one working day of loss.

10.20.3. If a security violation occurs, the contractor shall immediately report to the 309 AMARG Security Manager all available facts relating to each instance. The contractor shall take such precautions as the Security Manager may reasonably require for security purposes. The contractor shall take reasonable and prudent action to establish control of the scene, prevent further violation, and to preserve evidence until released by proper authority. The contractor shall cooperate fully and assist government personnel as the government conducts an investigation of the violation. Participation shall last until the investigation is completed.

10.21. CONTRACTOR TRAINING

10.21.1. The Contractor shall ensure employees complete required block training before performing any services. The Contractor will not be penalized for hours spent attending these required courses. The Contractor shall also ensure that annual training is completed before expiration date and that new mandatory training is completed within time frames designated by the Air Force.

10.21.2.

| COURSE NAME (EXAMPLE LIST)<br>(provided by Government except where noted) | NOTES |
|---|---|
| Foreign Object Damage (FOD) and Dropped Object (DO) Awareness | |
| Security Training Education and Motivation (phase I & II) | |
| Operations Security (OPSEC) | |

10.21.3. The Contractor shall ensure contractor personnel receive adequate training (safety, security, etc.). The Government will provide training on a space available/no charge basis when it is the only source of such instruction.

10.21.4. The Contractor shall maintain training and briefing records for all personnel attending Government furnished training and briefings, and provide the COR a roster of trained personnel upon request.

10.21.5. Mandatory Training. The Contractor shall ensure Contractor personnel attend at Contractor expense all applicable training required by law, Office of Personnel Management, Department of Defense (DoD), and/or AF policy (including command and local policy).

10.22. CONFLICTS OF INTEREST

10.22.1. The Contractor shall not assign to contract or task order (TO) performance any person, including any employee of the U.S. Government, if employing that person would create a conflict of interest under any law, regulation or policy of the United States Government. Additionally, the Contractor shall not assign to contract or TO performance, any person who is an employee of the Department of the Air Force, either military or civilian, unless such person seeks and receives approval according to DoD Directive 5500.7-R and any other applicable regulation or policy of the Department of the Air Force or its subordinate commands, as applicable, including but not limited to Air Force Materiel Command.

10.23. HOURS OF OPERATION

10.23.1. The contractor shall perform the services required under this contract during the following hours: Monday through Friday 0500-1700 – EXCEPT FEDERAL HOLIDAYS and other base closures. Work outside of these hours must be approved in advance by the COR.

10.24. FEDERAL HOLIDAYS

10.24.1. Unless otherwise required by the Government, the contractor shall not have access to the Government facilities on the following federal holidays: New Year's Day, Martin Luther King's Birthday, Washington's Birthday, Memorial Day, Independence Day, Juneteenth, Labor Day, Columbus Day, Veterans' Day, Thanksgiving Day, and Christmas Day. If a holiday falls on Saturday, it is observed on Friday and working hours are Monday through Wednesday for that week. If the holiday falls on a Sunday, it is observed on Monday. Any other circumstances that may prompt base closure, such as Days of National Mourning, may also restrict opportunity to perform the exchanges. In the event of a base closure, exchanges shall be performed on the next business day. There may also be up to six AFMC Goal Days and four Wingman Days per year awarded to Government civilian employees, as well as possible weather-related days and Government directed holidays. On these days, the contractor shall not generally be required to work and the Government shall not be billed for work not performed. NOTE: Good Friday is regarded as a holiday by the Arizona Wage Determination; however, Good Friday is not a Federal Holiday so the Government facilities will be open for work that day. The contractor shall plan on employees working business as usual on that day, unless the Government supervisor or section chief determines that such services are either unnecessary or will be less than a full normal workday.

10.25. DOWN DAYS/WEATHER DAYS

10.25.1. The Government will notify the contractor when official notice has been given for Energy/Weather Down days. The contractor is not expected to work on any Energy/Weather Down days.

10.26. CONSERVATION OF UTILITIES

10.26.1. The contractor shall instruct their employees in utilities conservation practices. The contractor shall be responsible for operating under conditions which prevent the waste of utilities.

10.27. PROJECT MANAGEMENT

10.27.1. The Contractor shall provide a Project Manager / Superintendent that will be knowledgeable on all projects they are assigned. The name of this person shall be designated in writing to the CO as part of the management plan provided with their proposal. The management plan will also include the experience and qualifications of planned resourced and management structure of personnel. After contract award the management plan will be updated including the IMS with construction phasing and operational safety plan IAW Appendix D (CDRL A004).

10.27.2. The Project Manager / Superintendent shall have at least three (3) years of experience in construction management for similar size / scope projects and be familiar with all aspects of general construction management, practices, methods, and materials.

10.27.3. The Project Manager / Superintendent shall have full authority to act on behalf of the Contractor on all contract matters relating to daily operation of this contract that do not have cost impacts.

10.27.4. The Project Manager / Superintendent shall be available by cellular telephone during normal duty hours.

10.27.5. Project or warranty related issues, the Project Manager / Superintendent shall respond as stated below:

10.27.5.1. For non-emergency issues, the Contractor / Subcontractor shall respond to the scene within twenty-four (24) hours of notification and begin repair actions.

10.27.5.2. For Emergency issues, the Contractor / Subcontractor shall respond to the scene within one (1) hour of notification and begin repair actions. Examples of emergency work are; work required to sustain / ensure continued AMARG mission operations, unscheduled water / electrical outages, no A/C or heat within a facility, imminent danger of Life/Health/Safety of personnel, roof leak directly impacting facility mission.

10.27.6. Sole Point of Responsibility. The Contractor's Project Manager / Superintendent shall act as the single point of accountability for all contract performance. The Prime Contractor is solely and fully responsible for the performance, management, and quality control of all subcontractors. Any delay, deficiency, or failure to perform by a subcontractor shall be the responsibility of the Prime Contractor to rectify immediately at no additional cost to the Government. A response from the Prime Contractor of "waiting for the subcontractor" to address any issue will be considered unacceptable, a failure to meet contractual obligations, and grounds for a formal Contract Discrepancy Report.

10.28. GOVERNMENT INSPECTIONS

10.28.1. The Contractor shall accommodate the inspection at the manufacturing location by Government officials prior to shipment.

10.28.2. The Contractor shall maintain an adequate inspection system and perform such inspections as will ensure that the work performed under the contract conforms to contract requirements. The Contractor shall maintain complete inspection records and make them available to the Government. All work shall be conducted under the general direction of the CO and is subject to Government inspection and test at all places and at all reasonable times before acceptance to ensure strict compliance with the terms of the contract. “Work” includes, but is not limited to, materials, workmanship, and manufacture and fabrication of components.

10.28.3. Government inspections and tests are for the sole benefit of the Government and do not:

10.28.3.1. Relieve the Contractor of responsibility for providing adequate quality control measures

10.28.3.2. Relieve the Contractor of responsibility for providing test and certification documentation as required in other paragraphs of the SOW

10.28.3.3. Relieve the Contractor of responsibility for damage to or loss of the material before acceptance

10.28.3.4. Constitute or imply acceptance

10.28.3.5. Affect the continuing rights of the Government after acceptance of the completed work.

10.28.4. The presence or absence of a Government inspector does not relieve the contractor from any contract requirement, nor is the inspector authorized to change any term or condition of the specification without the CO’s written authorization.

10.28.5. The Contractor shall, without charge, replace or correct work found by the Government not to conform to contract requirements, unless in the public interest the Government consents to accept the work with an appropriate adjustment in contract price. The Contractor shall promptly segregate and remove rejected material from the premises.

10.28.6. If the Contractor does not promptly replace or correct rejected work, the Government may:

10.28.6.1. By contract or otherwise, replace or correct the work and charge the cost to the contractor; or

10.28.6.2. Terminate for default the Contractor’s right to proceed.

10.28.7. Unless otherwise specified in the contract, the Government will accept, as promptly as practicable after completion and inspection, all work required by the contract or that portion of the work the CO determines can be accepted separately. Acceptance shall be final and conclusive except for latent defects, fraud, gross mistakes amounting to fraud, or the Government’s rights under any warranty or guarantee.

10.29. PROGRESS REPORT

10.29.1. The Contractor shall provide a status report (CDRL A005) on the second Monday of every month beginning the month after CO approval of the final design. The report shall document the status of the contractor effort towards meeting contract objectives including overall project status, status of scheduled milestones, activities accomplished, percent of planned work completed versus percent of time elapsed based on the IMS provided with identification of any problems/issues, and recommended solutions.

10.30. CONSTRUCTION ACTIVITIES

10.30.1. The Contractor shall provide all expertise, engineering and engineering support, labor, material, equipment, tools, testing, transportation, cleaning, waste removal and disposal, and supervision to accomplish the TO.

10.30.2. The Contractor and Subcontractors shall be in compliance with manufacturer's written instruction, applicable standards/specifications and IAW recognized industry practices.

10.31. FREE ZONE

10.31.1. A free zone boundary plan shall be created to delineate the boundary for construction access for security purposes. The free zone plan must be approved by the 355th Security Forces Squadron prior to use. All vehicles shall be operated by qualified individuals with adequate supervision and control. Privately owned vehicles (POV) shall not be allowed within the contractor free zone or on contractor access roads or haul roads. All privately owned vehicles driven by contractor personnel shall be parked at the staging area.

10.32. FENCING

10.32.1. The Contractor shall maintain free zone fencing throughout the duration of the contract. Prior to fence installation, the contractor shall submit the fencing plan and details for CORs approval. Upon completion of construction, all fence materials and barricades shall be disposed of outside the limits of Government- controlled lands.

10.33. PERMITS

10.33.1. The Contractor is responsible for federal, state, local, and other applicable permits, access (including off-base easements and leases), agreements, licenses, and certifications required to perform work. The Contractor shall maintain a library of these documents at the contractor’s site office.

10.33.2. Stormwater Pollution Prevention Plan (SWPPP) is required if the Contractor will disturb more than 1 acre. The SWPPP shall outline all of the activities and conditions at their site that could cause water pollution, and detail the steps the contractor will take to prevent the discharge of any unpermitted pollution and shall be submitted to the DMAFB Environmental office for approval.

10.34. BASE CIVIL ENGINEERING EXCAVATION PERMIT

10.34.1. At least 30 days prior to excavating, the Contractor shall obtain a template from the 355 CES Work Management. This form shall be filled out no less than 21 days prior to starting excavation work. The Contractor shall field mark the area of all intended excavations and alignment of new utility lines with flags or nonpermanent white paint.

10.35. WELDING, CUTTING AND BRAZING PERMITS

10.35.1. The Contractor will be required to attend the DMAFB Fire Department's Welding, Cutting, and Brazing certification class prior to performing any of these activities. Air Force (AF) Form 592 is required daily for all welding, cutting, brazing, soldering and similar hot work. The form shall be properly filled out and in possession of the performing these activities while all hot work is underway. Contact DMAFB Fire Department for details.

10.36. TEMPORARY AIRFIELD CONSTRUCTION WAIVER

10.36.1. A temporary construction waiver is required when one or more elements of a construction project violates criteria identified IAW UFC 3-260-01. The Contractor shall submit a request for waiver to the GPE with an FAA Form 7460- 1 "Notice of Proposed Construction or Alteration" at least 75 days prior to operation. Construction waiver shall only be planned for the duration of the construction project unless circumstances dictate otherwise.

10.37. FOREIGN OBJECT DAMAGE (FOD) PREVENTION

10.37.1. The Contractor shall sweep area and remove all FOD at the end of each work day. When FOD inadvertently leaves the free zone area the contractor shall address immediately IAW AFI 21- 101 and DMAFB / 309 AMARG requirements.

10.37.2. The Contractor shall keep free zone and access route concrete and paved surfaces clean and free of mud and debris. No amount of FOD will be tolerated or allowed.

10.37.3. Site Cleanliness and Foreign Object Debris (FOD) Control. The Contractor shall be solely responsible for maintaining a clean and orderly construction site at all times. The Contractor shall develop, submit, and implement a formal FOD prevention and control plan, to be approved by the COR prior to site mobilization. Daily cleanup of all FOD from the construction site and surrounding areas is mandatory. No migration of construction debris, dust, soil, or other materials from the designated construction site boundaries is permitted. The Prime Contractor shall be solely responsible for any and all damages, incidents, or mission impacts resulting from a failure to adhere to this requirement, and shall be responsible for the immediate cleanup of any material tracked onto adjacent roadways or properties at no additional cost to the Government.

10.38. WATER USE

10.38.1. Contractor shall coordinate use of water on DMAFB with Tucson Water. Contractor shall follow Tucson Water guidance, including utilization of fire hydrants, to include sufficient notice and use of backflow preventers.

10.39. CONSTRUCTION AND UTILITY OUTAGES

10.39.1. The Contractor shall submit an Air Force (AF) Form 103 to the GPE for all work including utility outages needed during the performance of the contract. Requests shall be made 28 calendar days prior to the anticipated outage date. The Contractor shall identify the specific utility system and/or circuits to be affected, location of work, time, date, duration, and contingency plan.

10.40. TEMPORARY UTILITIES

10.40.1. The Contractor will be allowed the use of common electrical and water utilities without reimbursement while performing work under this contract. Connection to utilities will be the responsibility of the contractor; connection and use are subject to review and approval by the GPE.

10.40.2. The Contractor shall be responsible to provide toilet facilities on the site for use by Contractor personnel.

10.41. MATERIALS AND WORKMANSHIP

10.41.1. All delivered equipment, materials, and products shall be new and of the grade for the intended purpose, unless otherwise specifically identified.

10.42. WEATHER PRECAUTIONS

10.42.1. Contractor shall ensure that weather sensitive materials are placed within the conditions recommended by the material supplier. No pavements shall be placed on frozen ground. Concrete placed when weather temperatures can be expected to fall below 32 degrees Fahrenheit, during cure, shall be covered with blankets as approved by the GPE.

10.43. STAGING AND STORAGE

10.43.1. If use of government premises is necessary, the Contractor shall confine those activities to locations approved by the GPE. Temporary structures are not permitted without advance approval by the GPE.

10.44. ENVIRONMENTAL CONTROLS

10.44.1. The Contractor shall ensure their employees are made aware that the performance of these services will occur in industrial areas. The industrial complex at Davis-Monthan AFB has the potential to expose workers to hazardous materials which may include, but is not limited to: hexavalent, chrome, cadmium, beryllium, lead, and others. While the Contractor shall not perform duties in a regulated area (where exposure are expected to exceed the permissible exposure limits set by 29 CFR1910.10xx), there may be contact hazards with these materials. The contractor shall ensure proper protective measures and training is taken to ensure contracted employees are protected from hazards. The 355th Air Base Wing (ABW) Bio-Environmental Engineering Flight produces BIO Survey of these areas which will be made available upon award.

10.44.2. The contractor shall be knowledgeable of and comply with all applicable Interstate, Federal, State, and Local laws, regulations, and requirements regarding environmental protection. In the event environmental laws/regulations change during the term of this contract, the contractor is required to comply as such laws come into effect.

10.44.3. If the contractor spills or releases any substance contained in 40 CFR 302 into the environment, the contractor or its agent shall immediately report the incident to the 309 AMAR Safety Office. The liability for the spill or release of such substances rests solely with the contractor and its agent.

10.45. SAFETY PROGRAM

10.45.1. The contractor shall implement a safety program that ensures protection of Government personnel and property. (See Appendix D)

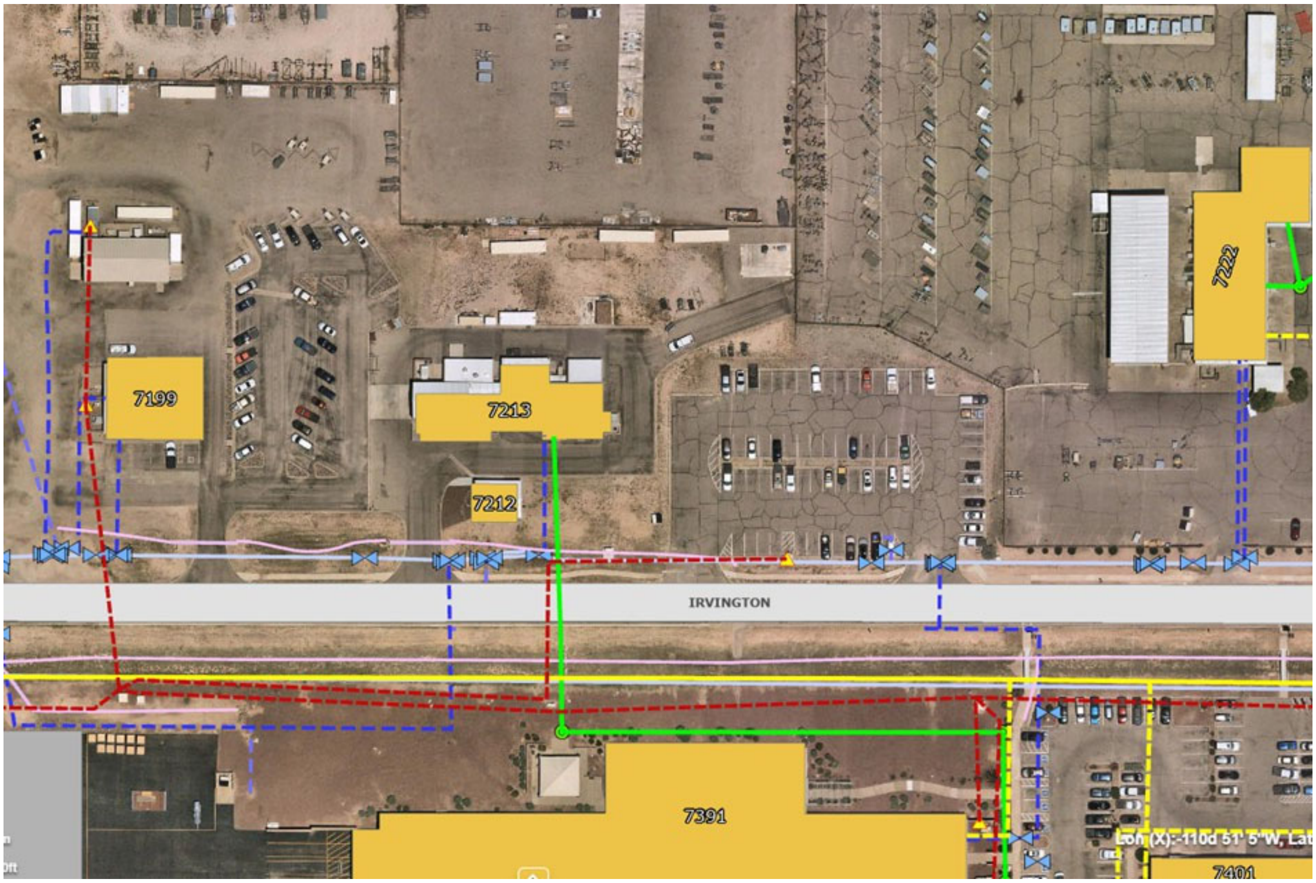
10.45.2. In the event of an accident/mishap, the contractor shall take reasonable and prudent action to establish control of the accident/mishap scene, prevent further damage to persons or property, and preserve evidence. An email copy of the mishap / incident notification shall be sent within three calendar days to the 309 AMARG Safety Office (CDRL A003).

10.46. GOVERNMENT OBSERVATIONS

10.46.1. During the period that the contractor is working on the Test Stands, Government personnel, other than CO, may, from time-to-time, observe contractor operations. However, these personnel may not interfere with contractor performance.

**END OF SOW**

# APPENDIX A – CONCEPTUAL SITE IMAGES




APPENDIX B – ACRONYMS AND ABBREVIATIONS

| | |
|---|---|
| ACS | Access Control System |
| ADA | Americans with Disabilities Act |
| AF | Air Force |
| AFB | Air Force Base |
| AFF | Above the Finished Floor |
| AFI | Air Force Instruction |
| AFMAN | Air Force Manual |
| AGE | Aerospace Ground Equipment |
| AMARG | Aircraft Maintenance and Regeneration Group |
| ANSI | American National Standards Institute |
| ASHRAE | American Society of Heating, Refrigerating and Air-Conditioning Engineers |
| ATFP | Anti-Terrorism and Force Protection |
| BACnet | Building Automation and Control Networks |
| CCTV | Closed-Circuit Television |
| CDRL | Contract Data Requirements List |
| CE | Civil Engineering |
| CER | Communications and Equipment Room |
| CFM | Cubic Feet per Minute |
| CFPS | Certified Fire Protection Specialist |
| CFR | Code of Federal Regulations |
| CLIN | Contract Line Item Number |
| CLP | City Light and Power |
| CO | Contracting Officer |
| COR | Contracting Officer’s Representative |
| COTS | Commercial-Off-The-Shelf |
| CRI | Color Rendering Index |
| DBIDS | Defense Biometric Identification System |
| DDC | Direct Digital Control |
| DOD | Department of Defense |
| EMCS | Energy Management Control System |
| FAA | Federal Aviation Administration |
| FACP | Fire Alarm Control Panels |
| FAR | Federal Aviation Regulation |
| FOC | Fiber Optic Cable |
| FOD | Foreign Object Damage |
| GFP | Government Furnished Property |
| GPE | Government Project Engineer |
| HDMI | High-Definition Multimedia Interface |
| HVAC | Heating Ventilation and Air Conditioning |
| IAW | In Accordance With |
| IBC | International Business Code |
| IDS | Intrusion Detection System |
| IES | Illuminating Engineering Society |
| IMC | International Mechanical Code |

| | |
|---|---|
| IPC | International Plumbing Code |
| IMS | Integrated Master Schedule |
| IT | Information Technology |
| LED | Light Emitting Diodes |
| LiDAR | Light Detection and Ranging |
| LPMC | Lighting Protection Master Certification |
| LPS | Lighting Protection System |
| LSHV | Low Speed, High Volume |
| Mil-Std | Military Standard |
| NEC | National Electric Code |
| NEMA | National Electrical Manufactures Association |
| NFPA | National Fire Protection Agency |
| NIPR | Non-Secure Internet Protocol Router |
| OO-ALC | Ogden Air Logistics Complex |
| OSHA | Occupational Safety and Health Administration |
| PoP | Period of Performance |
| POV | Privately Owned Vehicle |
| PPE | Personal Protective Equipment |
| PPMRB | Pre-Engineered Panelized Modular Relocatable Building |
| PSI | Pounds Per Square Inch |
| PSF | Pounds Per Square Foot |
| QASP | Quality Assurance Surveillance Plan |
| SOW | Statement Of Work |
| SSL | Solid State Luminaries |
| SWPPP | Storm Water Pollution Prevention Plan |
| TCIR/DART | Total Case Incident Rate/Days Away Restricted or Transferred |
| TO | Task Order |
| UFC | Unified Facilities Criteria |
| UFGS | Unified Facilities Guide Specifications |
| UL | Underwriters Laboratories |
| VAL/VER | Validate/Verify |
| VLFD | Vertical Lift Fabric Door |
| VOIP | Voice Over Internet Protocol |
| WBS | Work Breakdown Structure |

APPENDIX C – PUBLICATIONS/INSTRUCTIONS

AFIs, AFMANS, and AF Forms can be obtained from the internet website, https://www.e-publishing.af.mil. The Contractor will be responsible for researching and abiding by all applicable supplements not contained in the table below. All other references are available via general internet searches. It is the Contractors responsibility to ensure the most recent version is utilized when referenced.

| Regulation | Date | Reference | Description |
|---|---|---|---|
| AFI 13-213 | | | |
| AFI 13-213/Hill AFB Sup | | | |
| AFI 21-101 | | | |
| AFMAN 32-1065 | | Chapter 12 | Lightning Protection Systems |
| AFCEC | | All | Airfield and Helicopter Planning and Design; Pavement Design for Airfields; Dust Control for Roads, Airfields, and Adjacent Areas |
| AMARG Facility Design Standard | | | General Reference |
| ASHRE 62.1 | 2022 | All | Ventilation and Acceptable Indoor Air Quality |
| ASHRE 90.1 | 2022 | All | Energy Efficient Design of Low-Rise Residential Buildings |
| ASHRE 135 | 2020 | All | BACnet |
| ASHRE 189.1 | 2020 | All | Standard for the Design of High-Performance Green Buildings |
| DFARS | | 227.7102 | Commercial Items, Components, or Processes |
| DFARS | | 252.227-7015 | Technical Data – Commercial Items |
| DOD Directive 5500.7-R | | | |
| FAR 52.211-12 | | | |
| IBC | 2021 | 306.2 | Use and Occupancy Classification: Moderate-Hazard Factory Industrial, Group F-1 |
| IBC | 2021 | Chapter 6 | Types of Construction |
| IBC | 2021 | Table 506.3 | Frontage Increase |
| IBC | 2021 | 507.4 | Sprinklered, One-Story Buildings |
| IBC | 2021 | Table 508.4 | Required Separation of Occupancies |
| IMC | 2021 | | General Reference |
| IPC | 2021 | | General Reference |
| Hill AFB 21-100 | | | |
| MIL-DTL-85891 | | | |
| NEC | 2020 | | General Reference |
| NEMA 250 | 2020 | All | Enclosures for Electrical Equipment |
| NFPA 101 | 2020 | 40.1 & 40.2 | Means of Egress Requirements |
| NFPA 101 | 2020 | 40.2.8, 40.2.9, & 40.2.10 | Emergency Lighting Requirements |
| NFPA 409 | 2023 | Chapter 13 | Fire Suppression Installation and Testing |
| NFPA 780 | 2023 | 4.15 | Bonding of Metal Bodies |
| NFPA 780 | 2023 | Annex D | Inspection and Maintenance of LPS |
| OSHA 29 CFR 1910 | 2022 | 1910.36(h)(1) | Exit Routes and Emergency Planning |
| OSHA 29 CFR 1910 | 2009 | 1910.95 | Occupational Noise Exposure |

| | | | |
|---|---|---|---|
| OSHA 29 CFR 1910 | 2011 | 1910.147 | The Control of Hazardous Energy |
| OSHA 29 CFR 1910 | 1996 | 1910.196 | Compressed Gas and Compressed Air Equipment |
| T.O. 1-1-8 | | 2.12 | Engineered Media Blasting Removal Method |
| T.O. 1-1-8 | | 2.12.12 | Contamination Testing of Engineered Media |
| UFC 1-200-02 | 2022 | Chapter 2 | High Performance and Sustainable Building Requirements |
| UFC 3-201-01 | 2022 | 3-1.2 | Storm Drainage Systems |
| UFC 3-240-01 | | Table 3-3 | |
| UFC 3-260-01 | 2020 | All | Airfield and Heliport Planning and Design |
| UFC 3-301-01 | 2021 | 1609.1.2 | Aircraft Hangar Wind Loads |
| UFC 3-410-01 | 2021 | All | HVAC Systems |
| UFC 3-420-01 | 2021 | All | Plumbing Systems |
| UFC 3-520-01 | 2021 | 3-12 | Grounding, Bonding, and Static Protection |
| UFC 3-530-01 | 2023 | All | Interior and Exterior Lighting |
| UFC 3-570-01 | | | |
| UFC 3-575-01 | 2021 | Chapter 4 | Lighting and Static Electricity Protection Systems |
| UFC 3-601-2 | 2021 | All | Fire Protection Systems Inspection, Testing, and Maintenance |
| UFC 4-010-01 | 2022 | | |
| UFC 4-211-01 | 2021 | All | Aircraft Maintenance Hangars for defueled aircraft |
| UFC 4-211-01 | 2021 | 3-3.1.9 | Hangar Bay Egress |
| UFC 4-211-01 | 2021 | Chapter 3 & 5 | Electrical Power |
| UFC 4-211-01 | 2021 | 5-12.6.1 | Ventilation |
| UFC 3-301-01 | | 1609.1.2 | Aircraft Hangar Wind Loads |
| | | | |
| UFGS 08 34 16.10 | 2016 | All | Horizontal Rolling Steel Doors |
| UFGS 23 09 23.02 | 2019 | All | BACnet Direct Digital Control for HVAC and other Building Control Systems |
| UFGS 32 13 11 | | | |
| UFGS 32 13 14.13 | | | Concrete Paving for Airfields and Other Heavy Duty Pavements |

# APPENDIX D – SAFETY, FIRE PROTECTION AND HEALTH SPECIFICATION INDUSTRIAL SAFETY REQUIREMENTS

**SAFETY, FIRE PROTECTION AND HEALTH**
**SPECIFICATION**
**INDUSTRIAL SAFETY REQUIREMENTS**

**OGDEN AIR LOGISTICS COMPLEX**
**UNITED STATES AIR FORCE**
**HILL AIR FORCE BASE, UTAH 84056**

**LCBB**
**w/Enclosure**
**06 February 2026**
**Reviewed by:**
**AMARG/SE**
**520-228-**
**7310/8363**

## A. Safety Program Requirements

The contractor shall implement a safety program plan that ensures protection of Government personnel and property. **(CDRL A003)** The program shall consist of, as a minimum:

1. Mishap reporting, as defined in paragraph B1 below.

2. A Safety Plan that addresses, as a minimum, the subjects listed in Section II –Specific Requirements, and shall be used during the performance of the work described in the contract. The Safety Plan shall be reviewed by AMARG/SE (Safety Office) prior to commencement of any work described in this contract. (**CDRL A004**)

3. Routine and recurring surveillance to ensure the safety requirements of this contract are enforced.

4. Competent personnel to provide surveillance of the Safety Plan.

5. Identification of segregated work site locations for operations that cannot be co-mingled with general industrial operations and the process for ACO approval of operations and changes at these specific sites.

6. All contractor personnel shall be trained and qualified to perform their duties safely.

7. The contractor shall include a clause in all subcontracts requiring the subcontractors to comply with the safety provisions of this contract, as applicable.

**B. Mishap Notification**

1. The contractor shall notify AMARG/SE (520-228-7310/8363) or the AMARG Control (520-228-8777) after normal duty hours, and the designated Government Representative (GR), i.e., the ACO, PCO, or DCMA QAR (Quality Assurance Representative) within one (1) hour of all mishaps or incidents at or exceeding $2,000 (material + labor) in damage to DOD property entrusted by this contract, even if the government is wholly or partially reimbursed. This notification requirement shall also include physiological mishaps/incidents. A written or email copy of ALL the mishap/incident notification shall be sent within three calendar days to the COR, who will forward it to AMARG/SE. For information not available at the time of initial notification, the contractor shall provide the remaining information no later than 20 calendar days after the mishap, unless extended by the ACO. (**CDRL A003**)

Mishap notifications shall contain, as a minimum, the following information:

(a) Contract, Contract Number, Name and Title of Person(s) Reporting

(b) Date, Time and exact location of accident/incident

(c) Brief Narrative of accident/incident (Events leading to accident/incident)

(d) Cause of accident/incident, if known

(e) Estimated cost of accident/incident (material and labor to repair/replace)

(f) Nomenclature of equipment and personnel involved in accident/incident

(g) Corrective actions (taken or proposed)

(h) Other pertinent information

2. The contractor shall cooperate with any and all government mishap investigations. Additionally if requested by government personnel or designated government representative (GR), i.e., the ACO, PCO, or DCMA QAR (Quality Assurance Representative), the contractor shall immediately secure the mishap scene/damaged property and impound pertinent maintenance and training records, until released by safety investigators.

3. The contractor shall provide copies of contractor data related to mishaps, such as contractor analyses, test reports, summaries of investigations, etc. as necessary to support the government investigation.

4. The contractor shall support and comply with the safety investigation and reporting requirements of DAFI 91-204, Chapters 1 – 5.

## C. General Safety Requirements:

If the safety plan is modified, the contractor shall submit the proposed modification, in writing, to the Contract Administration Office safety representative.

The contractor is solely responsible for compliance with all federal, state and local laws, the Occupational, Safety and Health Act (OSHA) (Public Law 91-596) and the resulting standards, **OSHA Standards 29 CFR 1910 and 1926**, as applicable, Air Force Occupational Safety and Health Standards (AFOSH STD), to include AFMAN 91-203, *Air Force Occupational Safety, Fire, and Health Standards,* DAFI 91-204, *Safety Investigations and Reports*, *Contract Safety*, and the protection of their employees. Additionally, the contractor is responsible for the safety and health of all subcontractor employees.

The contractor shall ensure assigned personnel are adequately trained and qualified for the task being performed. Brief all personnel on the hazards involved with operations and applicable precautions to be taken. Should unidentified hazards arise, cease operations until actions are taken to eliminate or mitigate hazards to safe levels.

Compliance with OSHA and other applicable laws and regulations for the protection of contractor employees is exclusively the obligation of the contractor. **Note:** AFOSH STDs are annotated because many of the Air Force Standards exceed the OSHA standard criteria. If a conflict is noted, the most stringent requirement takes precedence. The government shall assume no liability or responsibility for the contractor's compliance or non-compliance with such requirements. The contractor shall furnish to each of his/her employees a place of employment, which is free from recognized hazards. The contractor shall brief his/her employees on the safety requirements of this contract and on hazards associated with prescribed tasks. The contractor is responsible for compliance with OSHA Public Law and the resultant standards identified within. In addition, the contractor is required to flow down the safety requirements/specification to all subcontractors. This applies to Federal Acquisition Regulation (FAR) 12 commercial acquisitions as well. This contract shall in no way require persons to work in surroundings or under working conditions which are unsafe or dangerous to their health. The contractor must coordinate and perform work so as not to impact the safety of government employees or cause damage to government property. This requires providing personnel with protective equipment and associated safety equipment as may be necessary. The contractor must also protect personnel from hazards generated by the work. If the contractor employs BILINGUAL speaking employees, they must post bilingual signs and have written procedures for specific tasks in applicable languages.

## SECTION II – SPECIFIC REQUIREMENTS

The contractor's prepared Safety Plan shall:

- Provide contact information for on-site personnel
- Include a work hazard analysis of the worksite and operations to be performed to include baseline hazard identification and required control measures
- Identify employee safety and health training requirements and the documentation process

The contractor shall ensure that each element identified below is adequately addressed in detail in the safety and health plan:

**<u>PEDESTRIAN CROSSWALKS:</u>** All contractor personnel are required to use the closest crosswalk, or traffic controlled intersection when crossing the road. Pedestrians must look both ways to ensure the coast is clear before stepping out into the crosswalk. Pedestrians <u>DO NOT</u> have the right of way unless they are already in the crosswalk. Contractor vehicle operators have the same responsibilities as pedestrians, to share the road and mutually observe and yield to pedestrians.

**<u>MOTOR VEHICLES</u>**: Contractor shall comply with the standards in*:* DoD Directive 5525.4, *Enforcement of State Traffic Laws on DoD Installations*", Nov 2, 1981, Para 3-4; DODI 6055.4*, DoD Traffic Safety Program,* 20 Jul 99; AFI 91-207, *USAF Traffic Safety Program,* 22 May 07; Each applies to all persons at any time on an Air Force Installation and includes all leased, owned, or privatized property including housing areas. In addition: AFI 13-213***, Airfield Management*,** 29 Jan 08, Paragraphs 1.3.6 and 4.4.2.1 applies to all contractors, sub-contractors, vendors, commercial delivery companies, and all other private business vehicles who operate anywhere on Hill Air Force Base, including the airfield (to include the industrial areas and any buildings or hangars located upon the airfield) in support of their mission.

**<u>HOUSEKEEPING</u>:** Housekeeping shall be conducted according to the requirements in OSHA Standard 29 CFR 1910.141. ***CLEAN AS YOU GO*** will be enforced.

APPENDIX E – UFC 4-211-02 Figure 2-5, Clean-Dirty Schematic